

CME 296: Diffusion & Large Vision Models



Afshine Amidi & Shervine Amidi

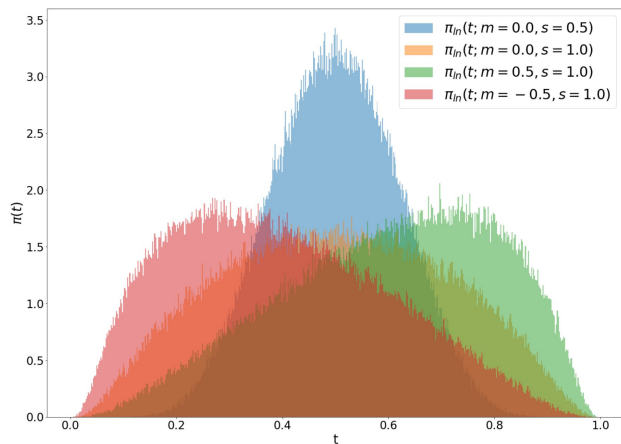


Recap of last episode...

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_1, x_0} \left[\|u_{\theta}(x_t, t) - (x_1 - x_0)\|^2 \right]$$

Recap of last episode...

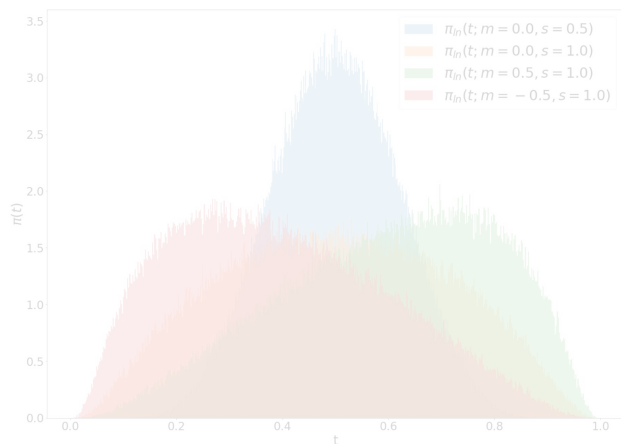
$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_1, x_0} \left[\|u_{\theta}(x_t, t) - (x_1 - x_0)\|^2 \right]$$



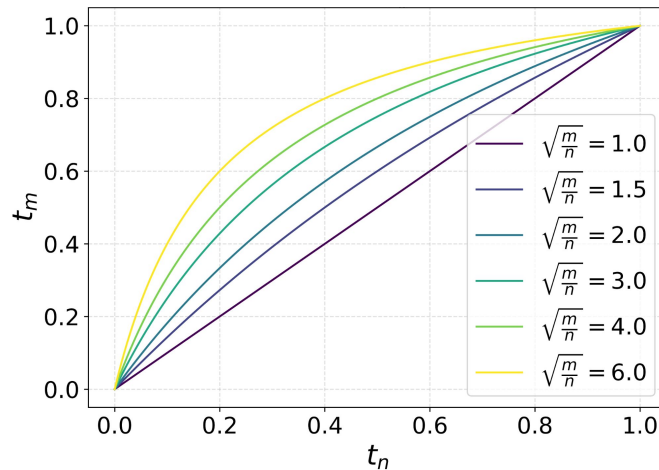
Logit-Normal distribution

Recap of last episode...

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_1, x_0} \left[\|u_{\theta}(x_t, t) - (x_1 - x_0)\|^2 \right]$$



Logit-Normal distribution



$$t_m = \frac{\sqrt{\frac{m}{n}} t_n}{1 + \left(\sqrt{\frac{m}{n}} - 1\right) t_n}$$

Recap of last episode...

1

Pre-training



Learn **how** to
generate images

Recap of last episode...

1

Pre-training



Learn **how** to
generate images

2

Post-training



Learn how to
generate **good**
images

Recap of last episode...

1

Pre-training



Learn **how** to
generate images

2

Post-training



Learn how to
generate **good**
images

3

Tuning



Learn how to
generate images
for a special case

Recap of last episode...

1

Pre-training



Learn **how** to
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Post-training



Learn how to
generate **good**
images

3

Tuning



Learn how to
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for a special case

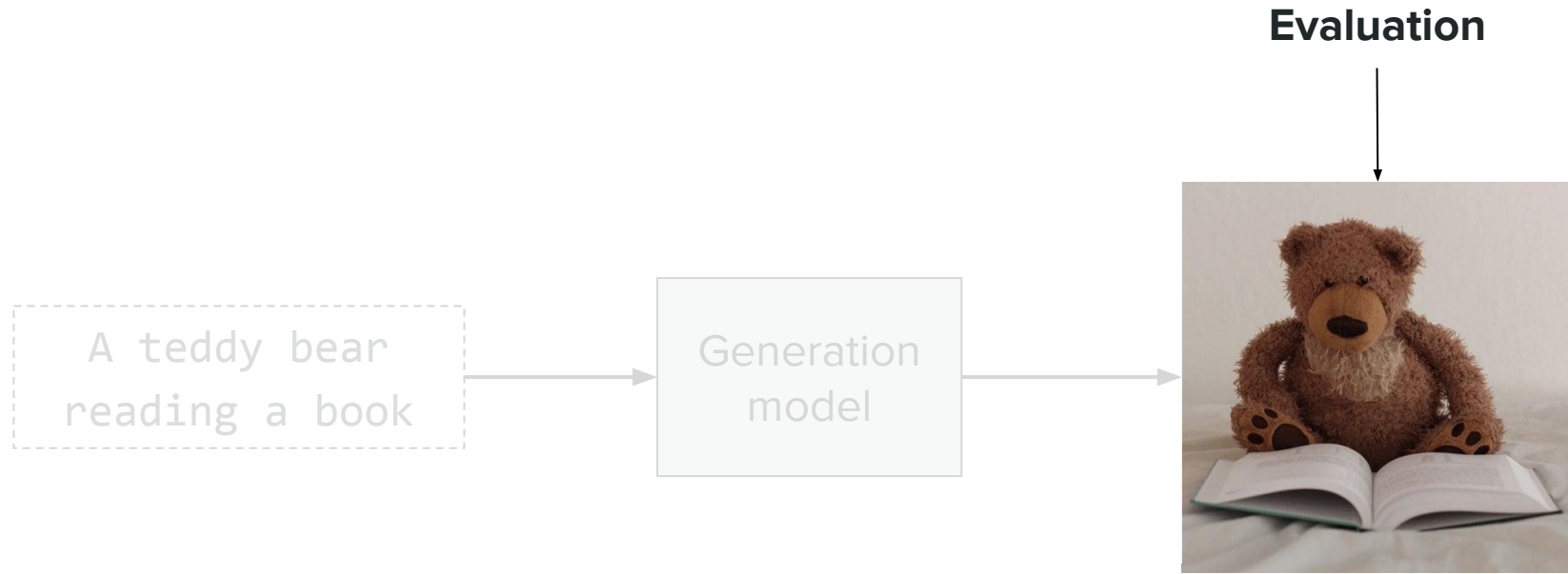
4

Distillation



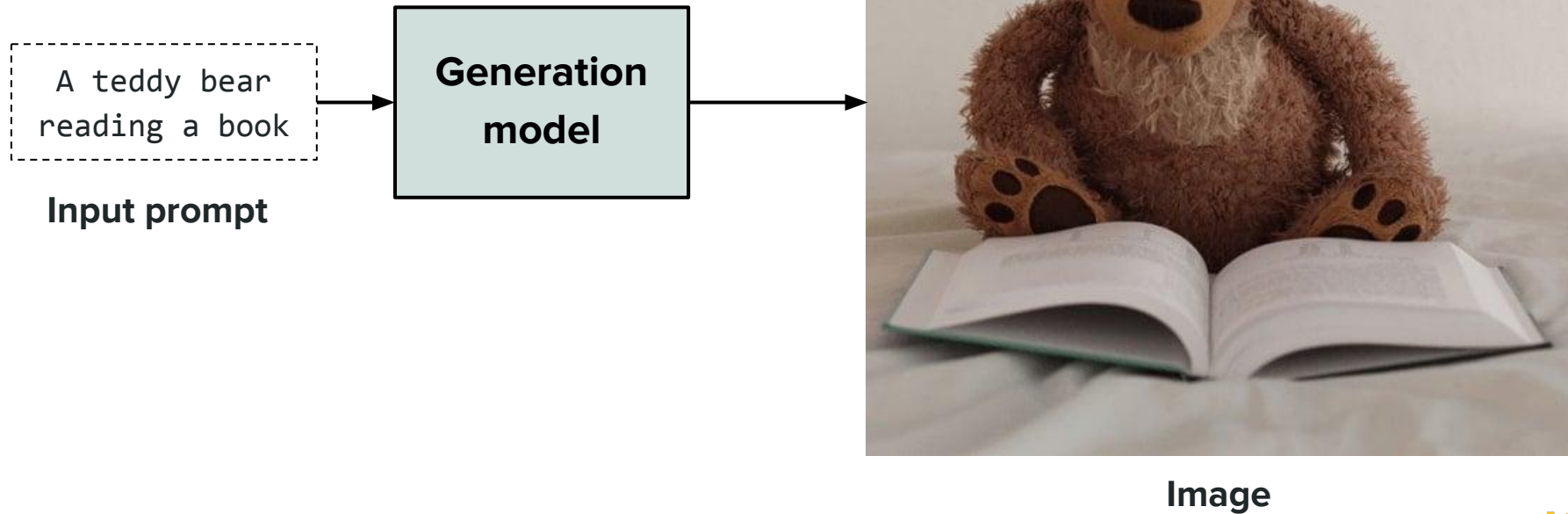
Learn how to
generate images,
fast

Today's lecture

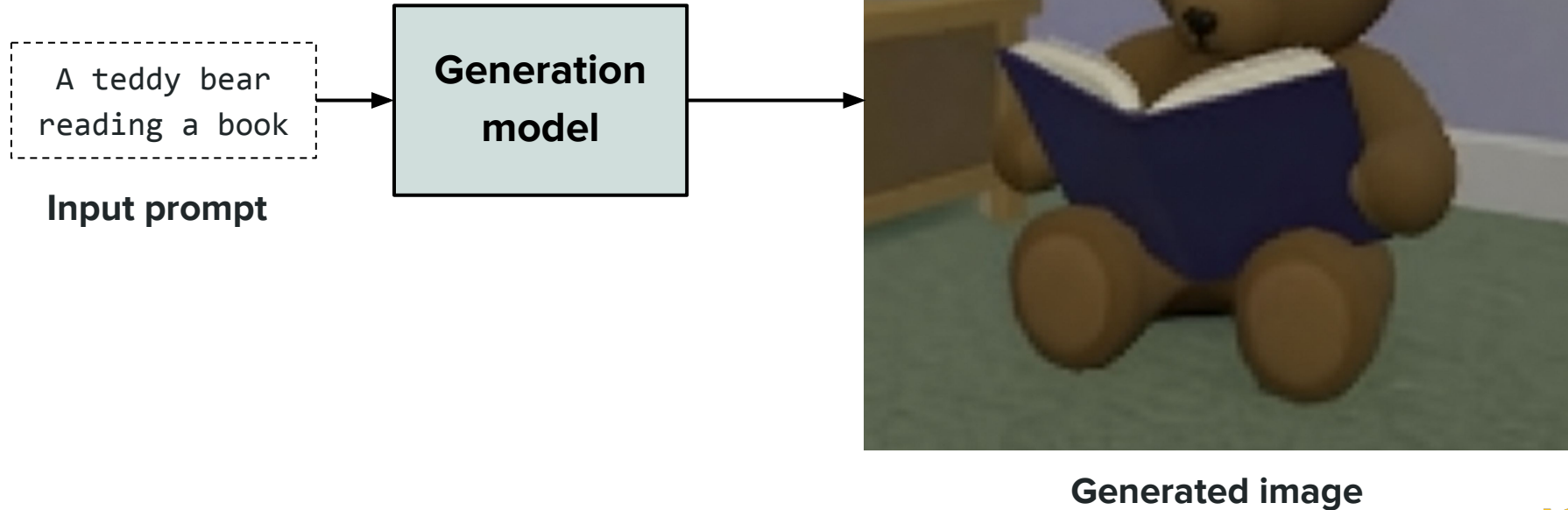


Today's lecture: **Evaluation**

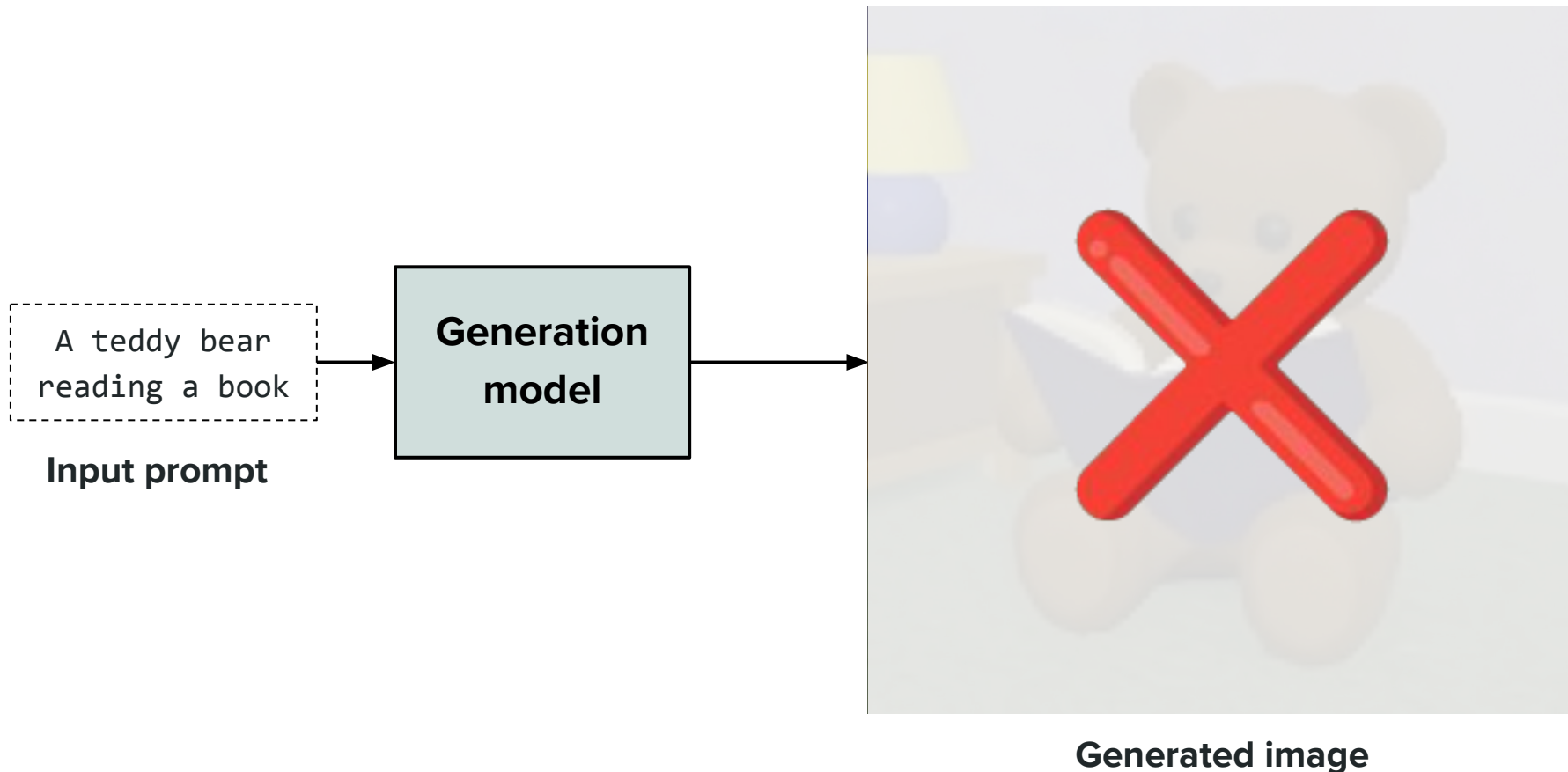
Setup



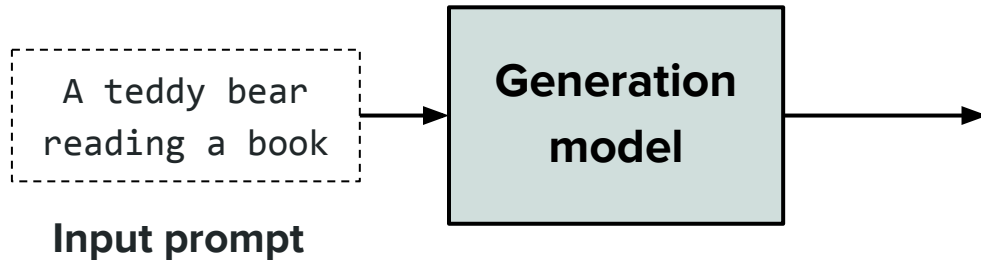
Motivating example



Motivating example: does not look real / high quality!

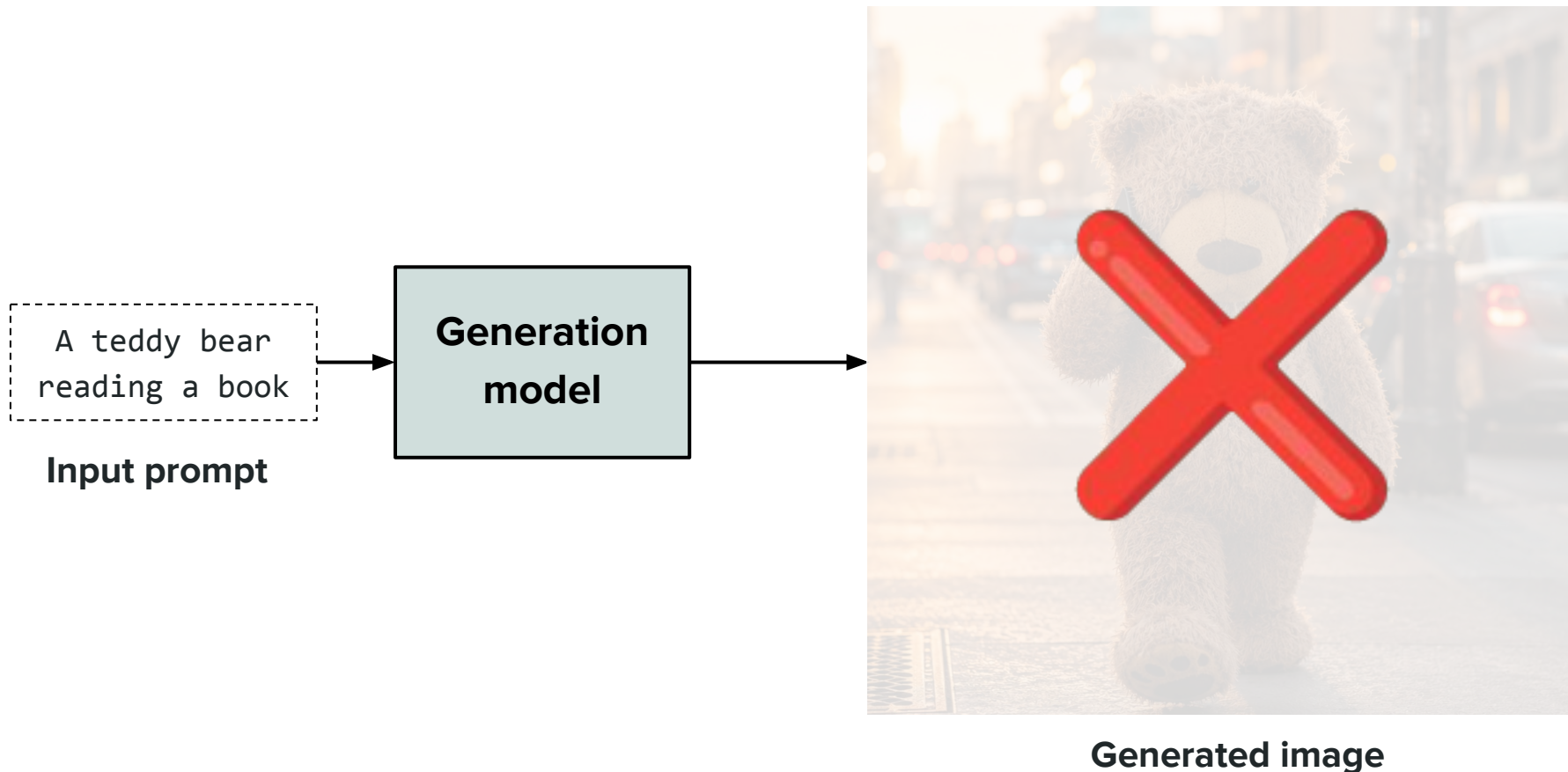


Motivating example

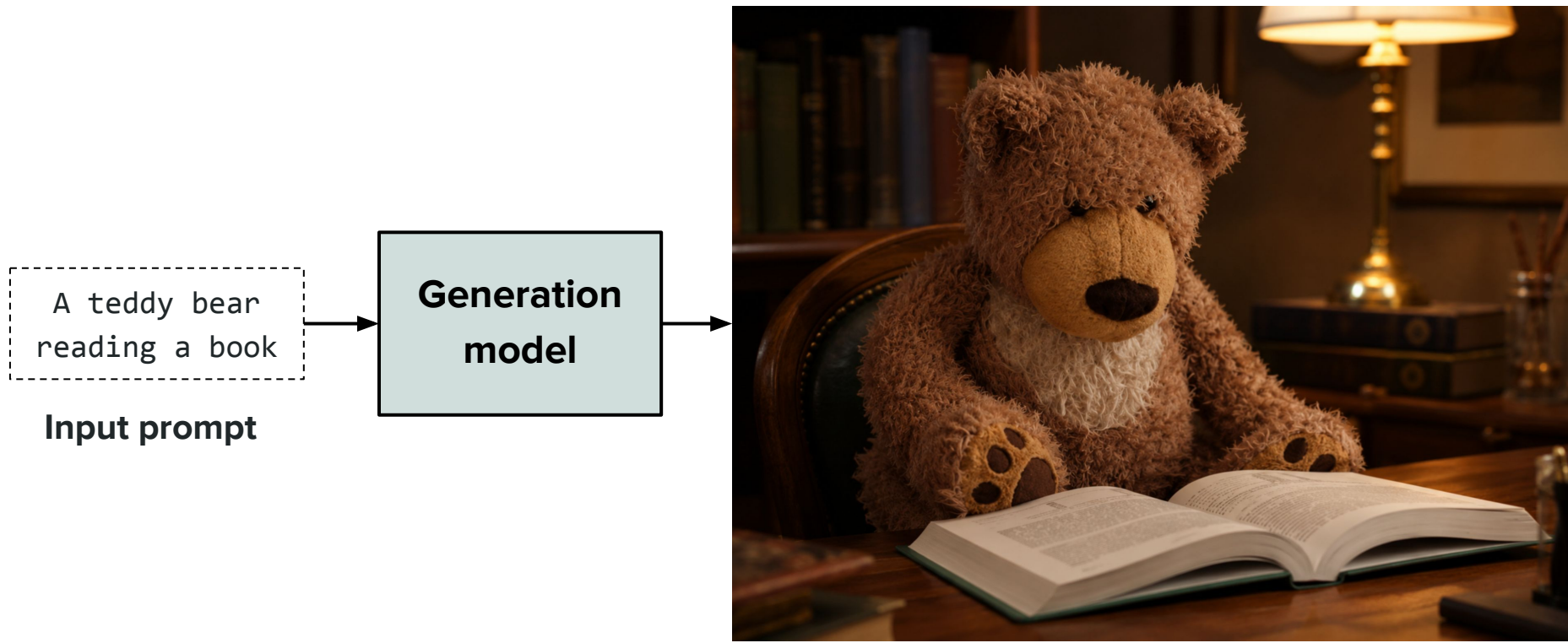


Generated image

Motivating example: does not follow prompt!

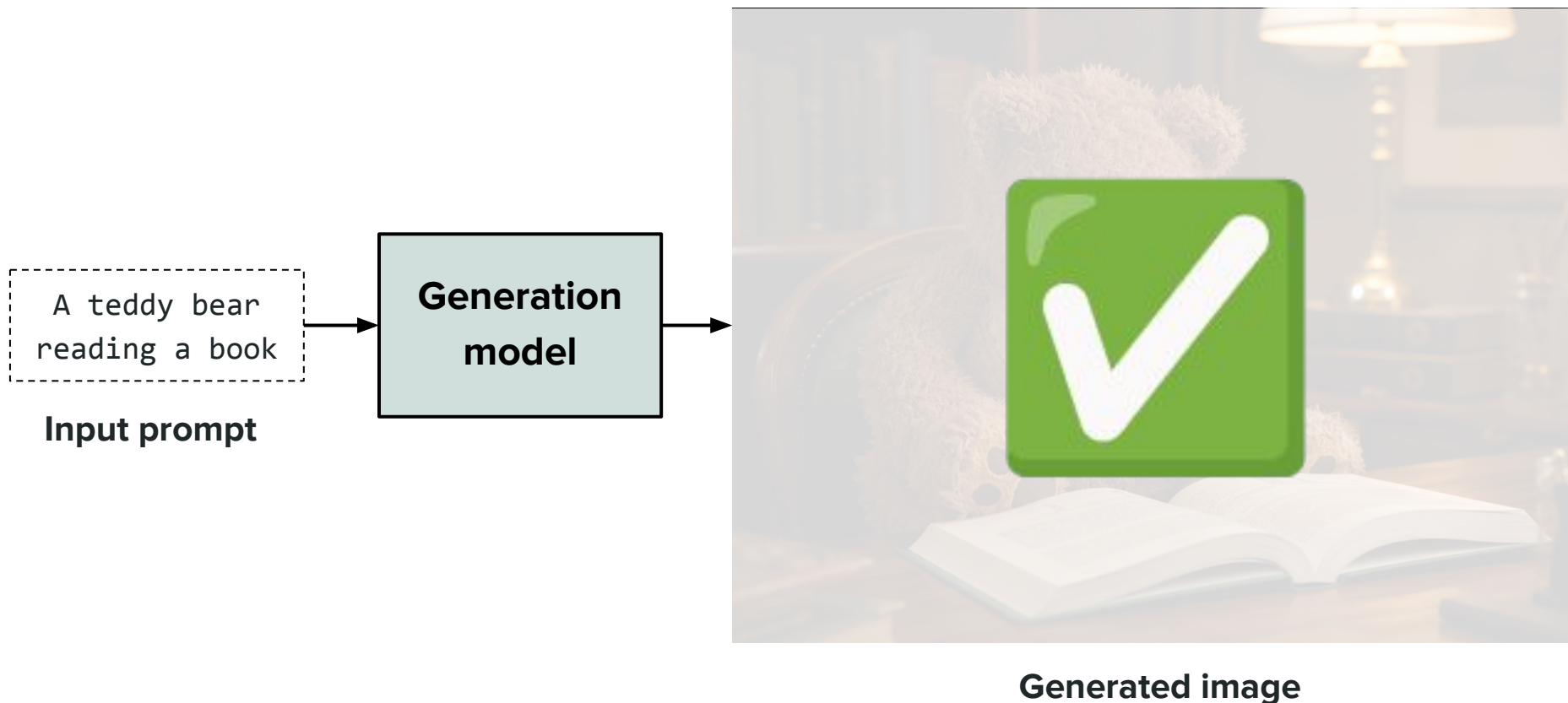


Motivating example



Generated image

Motivating example: follows prompt and high quality!



Tractable scope

Aesthetics

"Is this a good picture?"

- Physical plausibility
- Cleanliness
- Perceptual quality
- Realism



Tractable scope

Aesthetics

"Is this a good picture?"

- Physical plausibility
- Cleanliness
- Perceptual quality
- Realism



Prompt adherence

"Did it follow instructions?"

- Object recall
- Counting
- Text rendering
- Style adherence

A teddy bear
reading a book



...but not exhaustive!

Other notable dimensions.

-  Safety
-  Diversity
-  Memorization
-  Bias



Diffusion & Large Vision Models

Human ratings

Reference-free metrics

Reference-based metrics

Multimodal LLMs

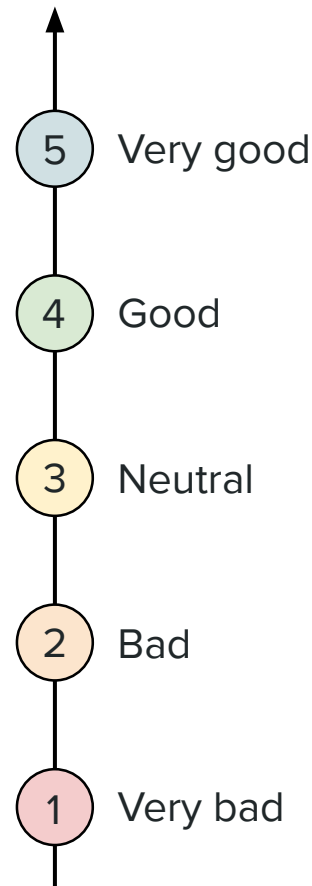
MLLM-as-a-Judge

Benchmarks

Attempt 1: rate images on a scale

A teddy bear reading a book

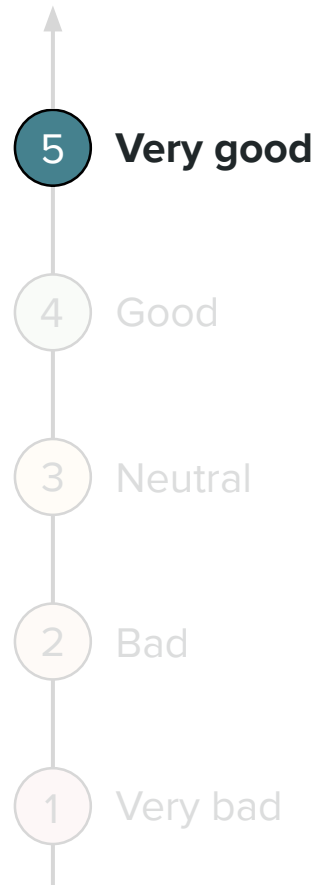
Model A



Attempt 1: rate images on a scale

A teddy bear reading a book

Model A



Attempt 1: rate images on a scale

$$\text{Score} = \frac{\text{Sum of ratings}}{\text{Number of ratings}}$$

Average of ratings

Discussion.

- **Good** to have **nuance** in score
- Ratings may be **noisy**
- **Hard** to rate on an **absolute** scale

Attempt 2: rate images on a **binary** scale

A teddy bear reading a book

Model A



1

Good

or

0

Bad

Attempt 2: rate images on a **binary** scale

A teddy bear reading a book

Model A



1

Good

or

0

Bad

Attempt 2: rate images on a **binary** scale

$$\text{Score} = \frac{\text{Sum of ratings}}{\text{Number of ratings}}$$

Proportion of pass

Discussion.

- **Easier** task compared to nuanced scale
- Still **hard** to rate on an **absolute** scale

Attempt 3: rate images via pairwise comparisons

A teddy bear reading a book

Model A



Model B



>
=
<

Attempt 3: rate images via pairwise comparisons

A teddy bear reading a book

Model A



Model B



>

=

<

Win rate

$$\text{Score} = \frac{\text{Number of wins}}{\text{Number of comparisons}}$$

Win rate

Discussion.

- **Much easier** task than absolute scale!
- Win rate **depends on** who we are comparing with who...

Cases where win rate is not great...

■ Artificial Analysis Text to Image Leaderboard Artificial Analysis

Added to the leaderboard in the last month:

HiDream-O1-Image, HiDream-O1-Image-Dev, HiDream-O1-Image-Dev-2604, GPT Image 2 (high), MAI-Image-2

Category: All ▾		Current models	All models	All	Open weights	First party foundation models	All	Global Leaderboard	Personal Leaderboard
↑↓	Range ⓘ	Creator ↑↓	Model ↑↓			Samples ↑↓		Released ↑↓	API Pricing ¹ ↑↓
1	1	🌐 OpenAI	GPT Image 2 (high)			9,774		Apr 2026	\$211.0 /1k imgs
2	2-3	🌐 OpenAI	GPT Image 1.5 (high) ⓘ			5,335		Dec 2025	\$133.0 /1k imgs
3	2-3	🇸🇩 Google	Nano Banana 2 (Gemini 3.1 Flash Image Preview)			7,113		Feb 2026	\$67.0 /1k imgs
4	4	🇸🇩 Google	Nano Banana Pro (Gemini 3 Pro Image)			4,610		Nov 2025	\$134.0 /1k imgs
5	5-7	🇺🇸 Microsoft	MAI-Image-2			4,240		Mar 2026	\$50.0 /1k imgs
6	5-7	🇨🇳 ByteDance Seed	Seedream 4.0			10,649		Sept 2025	\$30.0 /1k imgs
7	5-9	🇩🇪 Black Forest Labs	FLUX.2 [max]			4,606		Dec 2025	\$70.0 /1k imgs

...and intuition behind how to fix this!

Objective. Compute win rate... by **taking into account "strength" of opponent!**

Model A



Bad model



Sure, Model A won. **So what?** 🙋

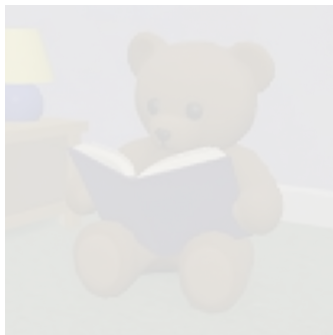
...and intuition behind how to fix this!

Objective. Compute win rate... by **taking into account "strength" of opponent!**

Model A



Bad model



Model A



Good model



Sure, Model A won. **So what?** 🙄

WOW, Model A won!!!! 🎉

Elo score

Model A $\longleftrightarrow R_A = 1000$ **Bad model** $\longleftrightarrow R_B = 600$

- 1 Compute **expected** score

$$E_A = \frac{1}{1 + 10^{\frac{R_B - R_A}{400}}} \approx 0.9$$

Elo score

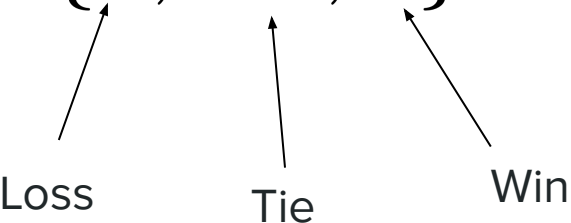
Model A $\longleftrightarrow R_A = 1000$

Bad model $\longleftrightarrow R_B = 600$

- 1 Compute **expected** score
- 2 Obtain **actual** score

$$S_A \in \{0, 0.5, 1\}$$

Loss Tie Win



Elo score

Model A $\longleftrightarrow R_A = 1000$

Bad model $\longleftrightarrow R_B = 600$

- 1 Compute **expected** score
- 2 Obtain **actual** score
- 3 **Compare** actual score with expected score

$$\Delta = S_A - E_A \quad \left\{ \begin{array}{ll} \Delta = +0.1 & \text{if Model A wins} \\ \Delta = -0.4 & \text{if there is a tie} \\ \Delta = -0.9 & \text{if Model A loses} \end{array} \right.$$

Elo score

Model A $\longleftrightarrow R_A = 1000$

Bad model $\longleftrightarrow R_B = 600$

- 1 Compute **expected** score
- 2 Obtain **actual** score
- 3 **Compare** actual score with expected score
- 4 **Update** rating

$$R_A \leftarrow R_A + K \times \Delta$$

Leaderboard of text-to-image models

■ Artificial Analysis Text to Image Leaderboard Artificial Analysis

Added to the leaderboard in the last month:

HiDream-O1-Image, HiDream-O1-Image-Dev, HiDream-O1-Image-Dev-2604, GPT Image 2 (high), MAI-Image-2

Category: All ▾									
Current models All All weights First party foundation models All Global Leaderboard Personal Leaderboard									
↑↓	Range ⓘ	Creator ↑↓	Model ↑↓	Elo ↑↓	95% CI	Samples ↑↓	Released ↑↓	API Pricing ¹ ↑↓	
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Limitations with human ratings

-  Expensive
-  Slow
-  Subjective
-  Not necessarily ground truth (fatigue, cultural bias)



Diffusion & Large Vision Models

Human ratings

Reference-free metrics

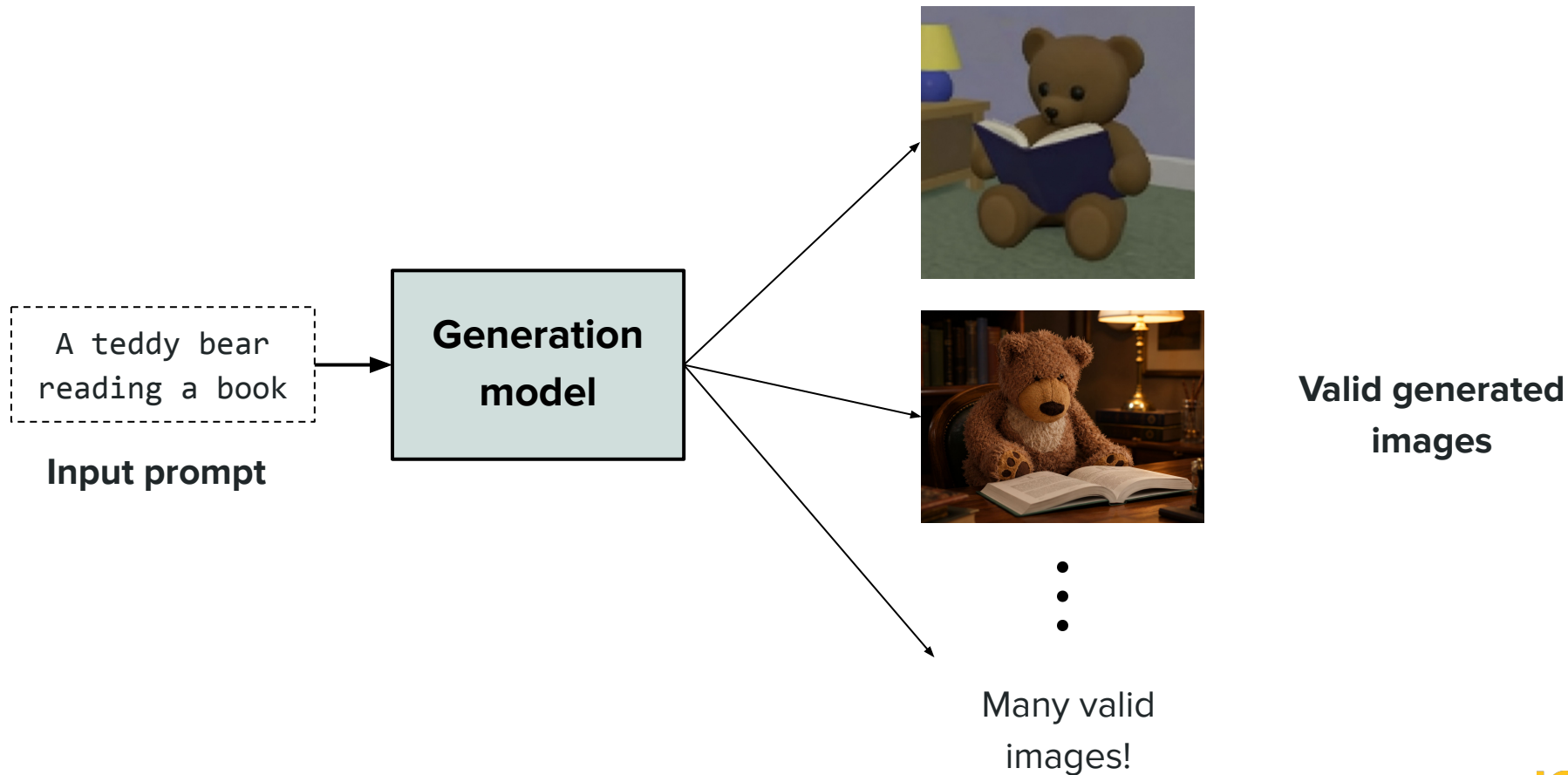
Reference-based metrics

Multimodal LLMs

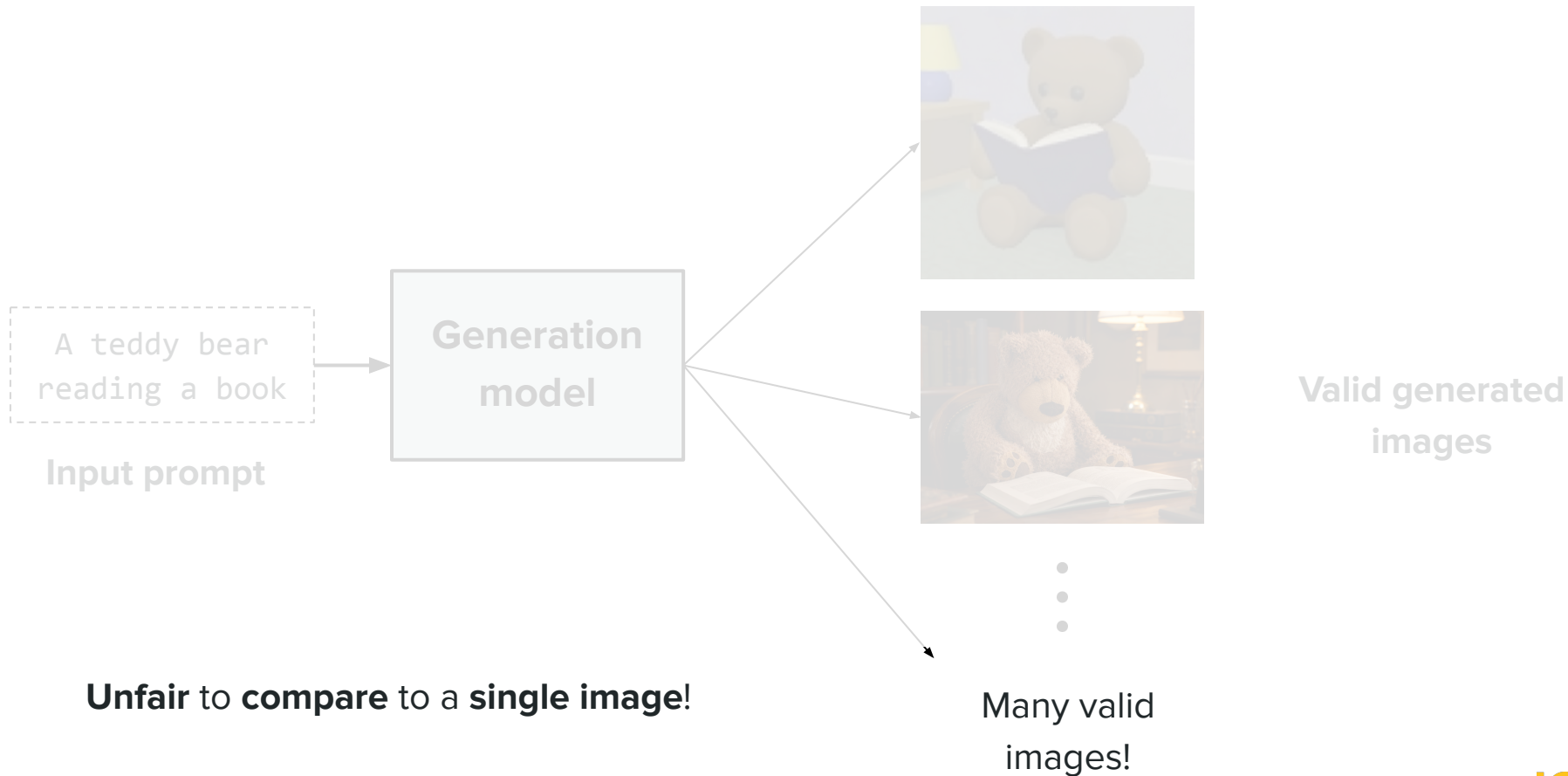
MLLM-as-a-Judge

Benchmarks

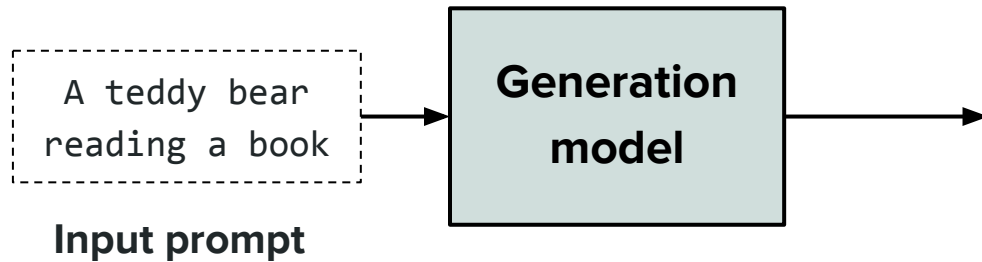
Motivation



Motivation



Back to our problem



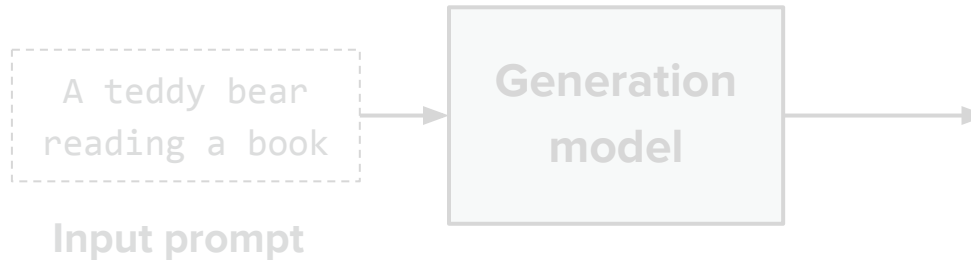
Input prompt



Generated image

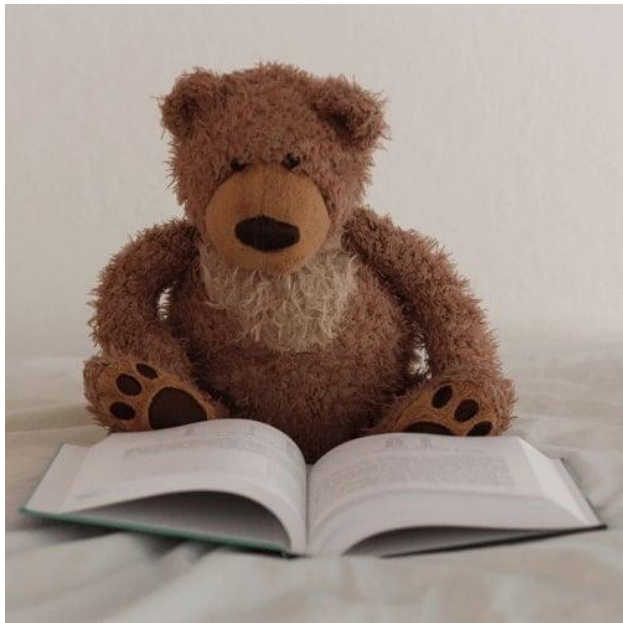
Back to our problem

1 Aesthetics



Generated image

Aesthetics

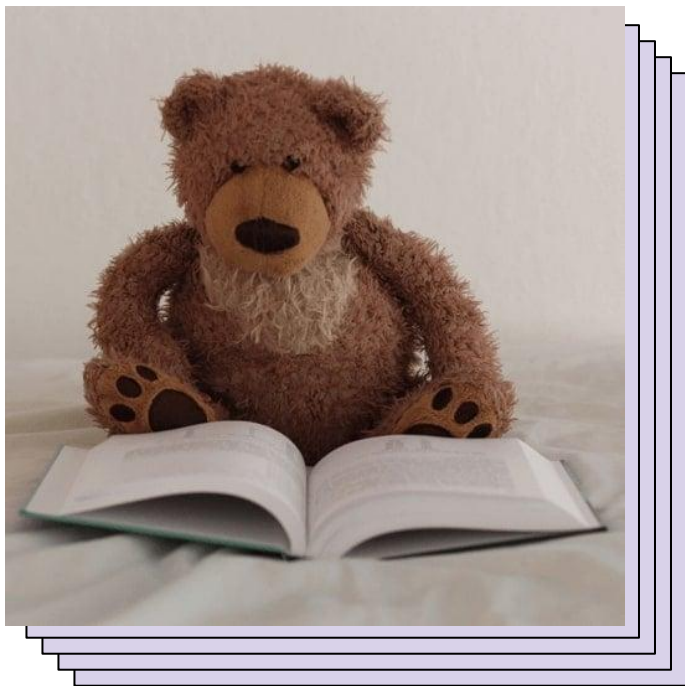


Real image



Generated image

Aesthetics

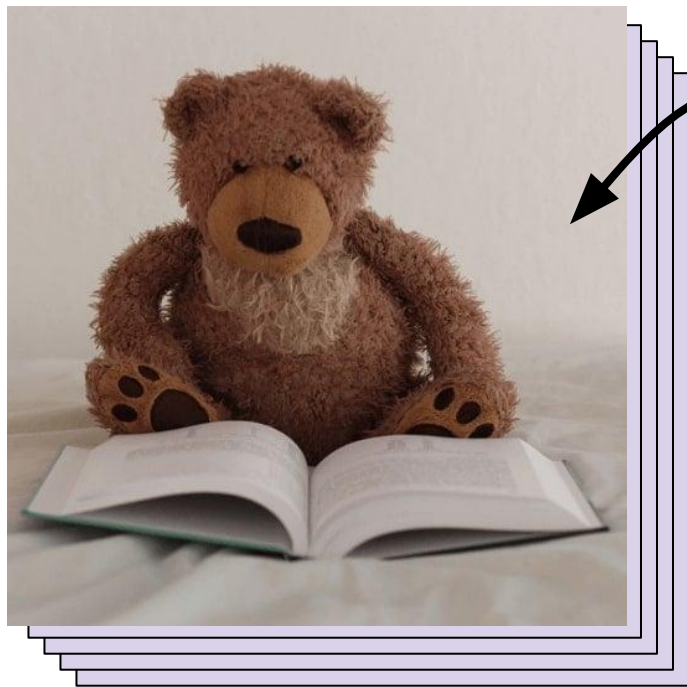


Real image **distribution**



Generated image **distribution**

Aesthetics



Real image **distribution**

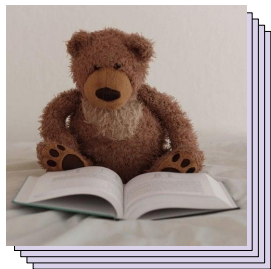
?



Generated image **distribution**

Aesthetics

Real image distribution

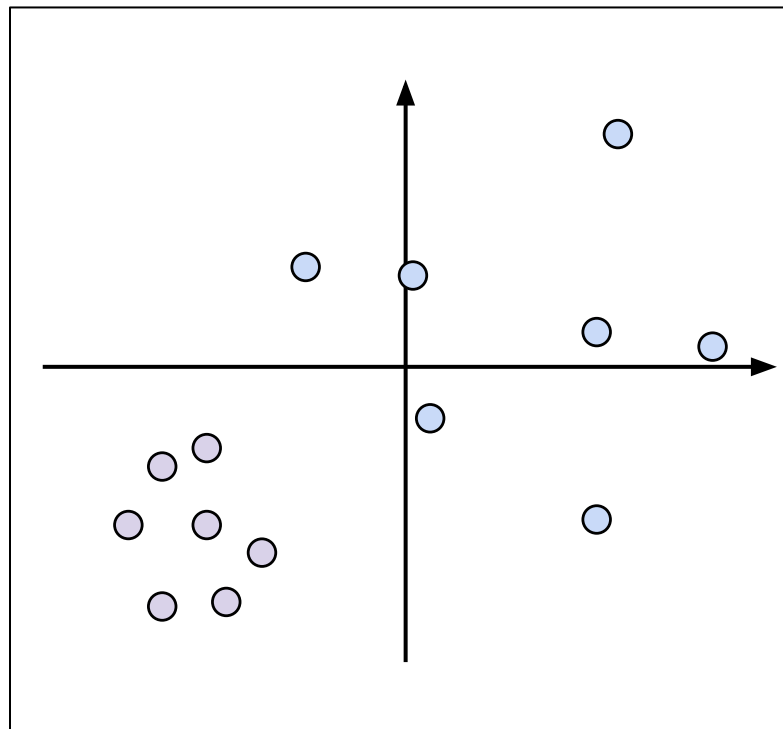


Pre-trained
encoder

Generated image distribution

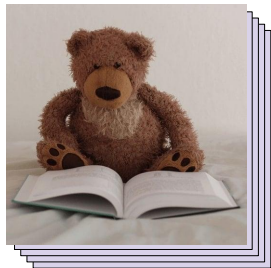


Pre-trained
encoder

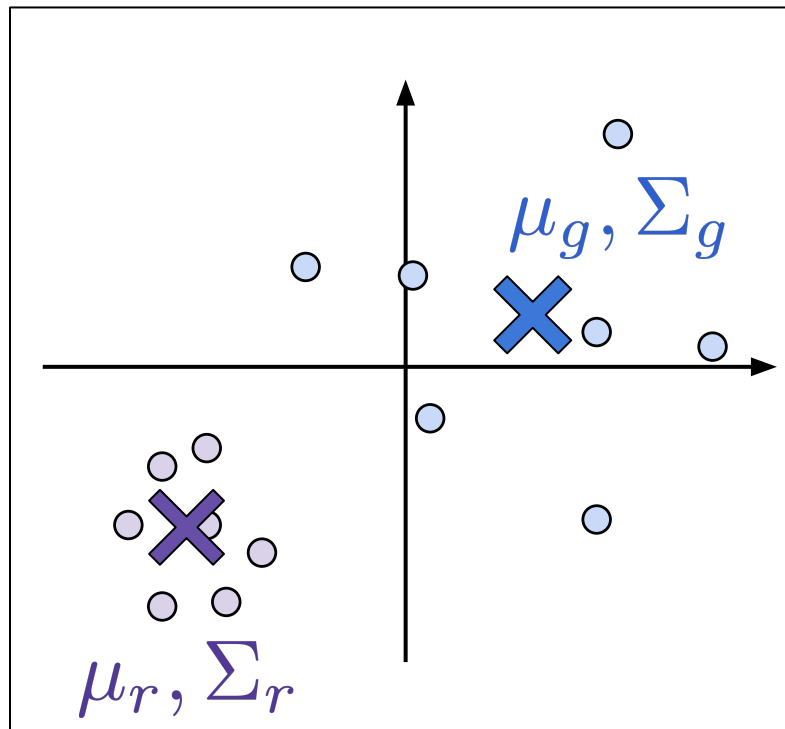


Aesthetics

Real image distribution



Pre-trained
encoder



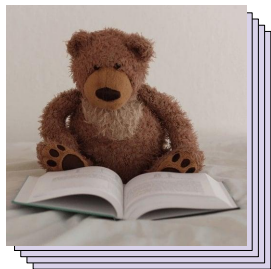
Generated image distribution



Pre-trained
encoder

Aesthetics

Real image distribution

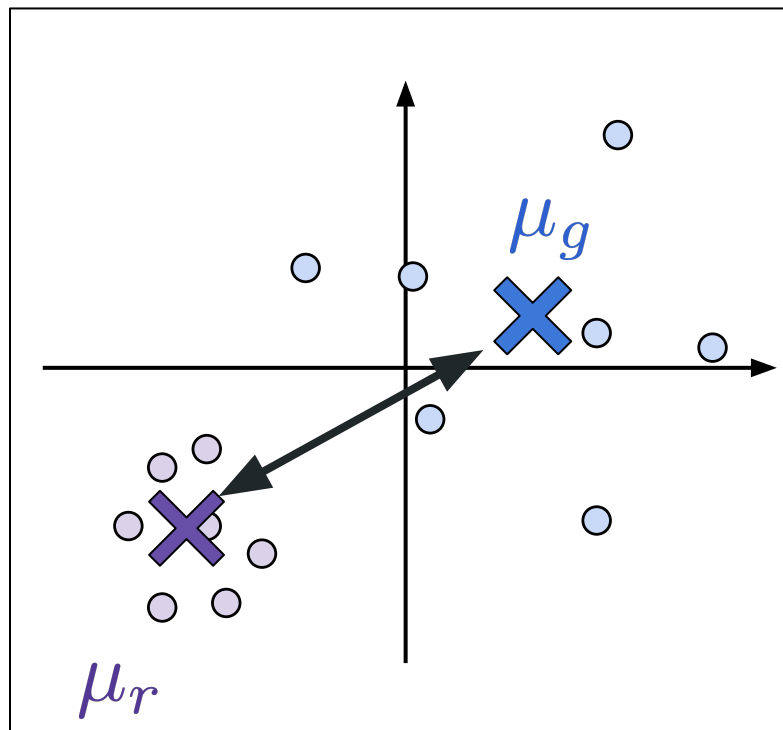


Pre-trained
encoder

Generated image distribution

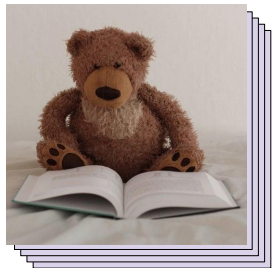


Pre-trained
encoder

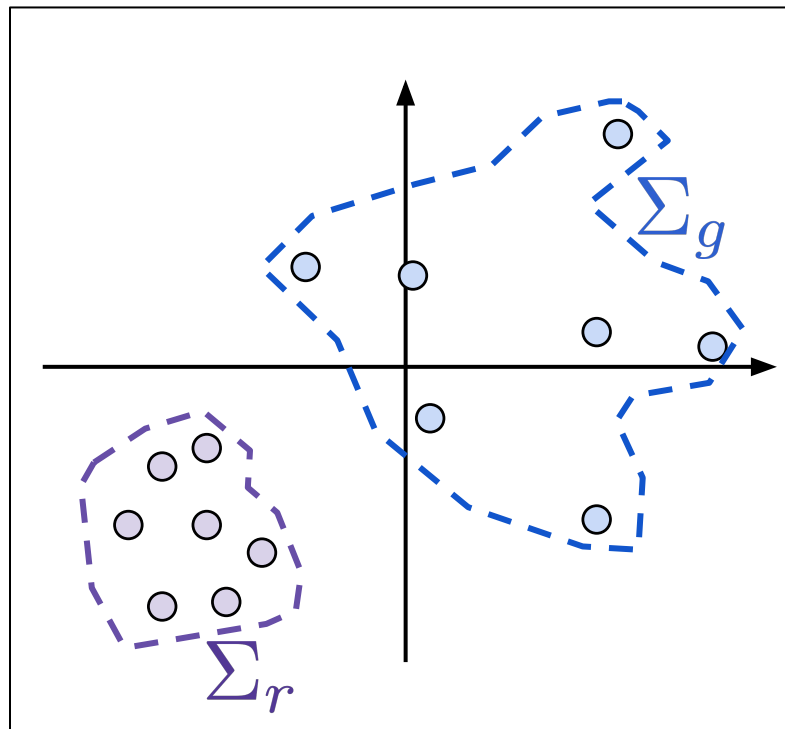


Aesthetics

Real image distribution



Pre-trained
encoder



Generated image distribution



Pre-trained
encoder

Distance between real and generated distributions

Idea. Quantify distance between generated and real images

$$\underbrace{\|\mu_r - \mu_g\|^2}_{\text{Location difference}} + \underbrace{\text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right)}_{\text{Shape difference}}$$

Fréchet inception distance

FID = **F**réchet **I**nception **D**istance

Idea. Quantify distance between generated and real images

$$\text{FID} = \underbrace{\|\mu_r - \mu_g\|^2}_{\text{Location difference}} + \underbrace{\text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right)}_{\text{Shape difference}}$$

The **lower**, the **better**!

We already saw plots using FID in the class!

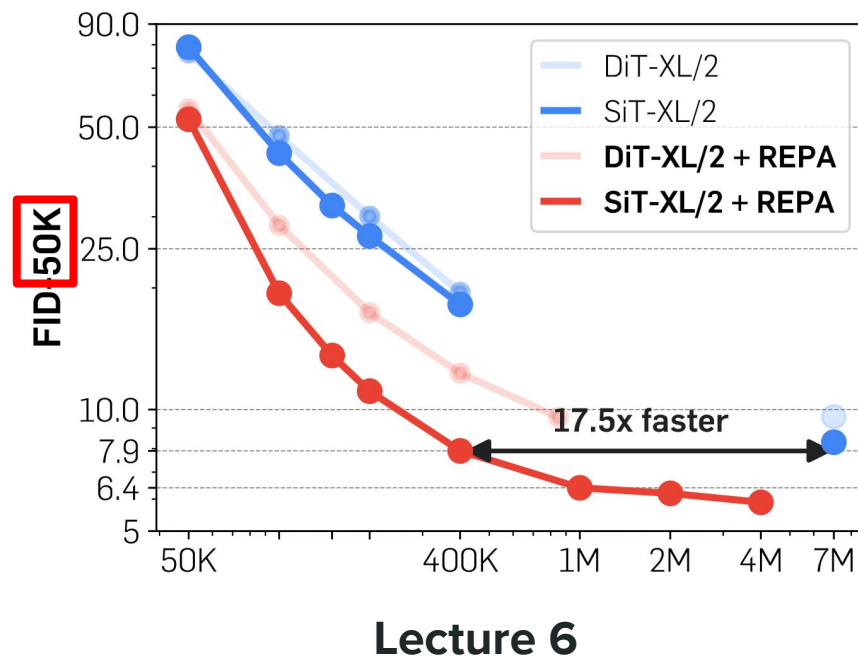
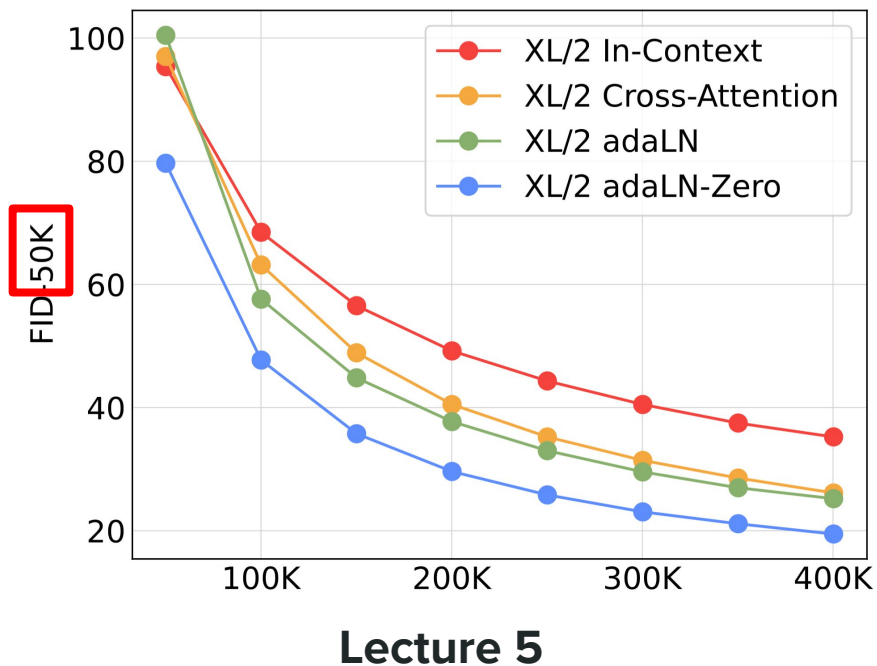


Figure from *Scalable Diffusion Models with Transformers*, Peebles et al., 2022.

Figure from *Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think*, Yu et al., 2024.

Discussion on the FID

What it measures.

- **Location difference.** Quality, style
- **Shape difference.** Diversity, variety

Discussion on the FID

What it measures.

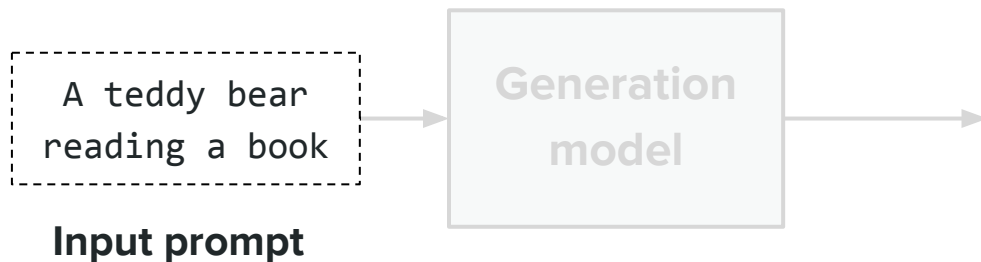
- **Location difference.** Quality, style
- **Shape difference.** Diversity, variety

Discussion. Depends on:

- **Sample size** (common: FID-50k)
- **Reference** data distribution
- **Normality** assumption

Back to our problem

- 1 Aesthetics
- 2 **Prompt adherence**



Generated image

Reminder from Lecture 4: CLIP!

CLIP = **C**ontrastive **L**anguage-**I**mage **P**retraining

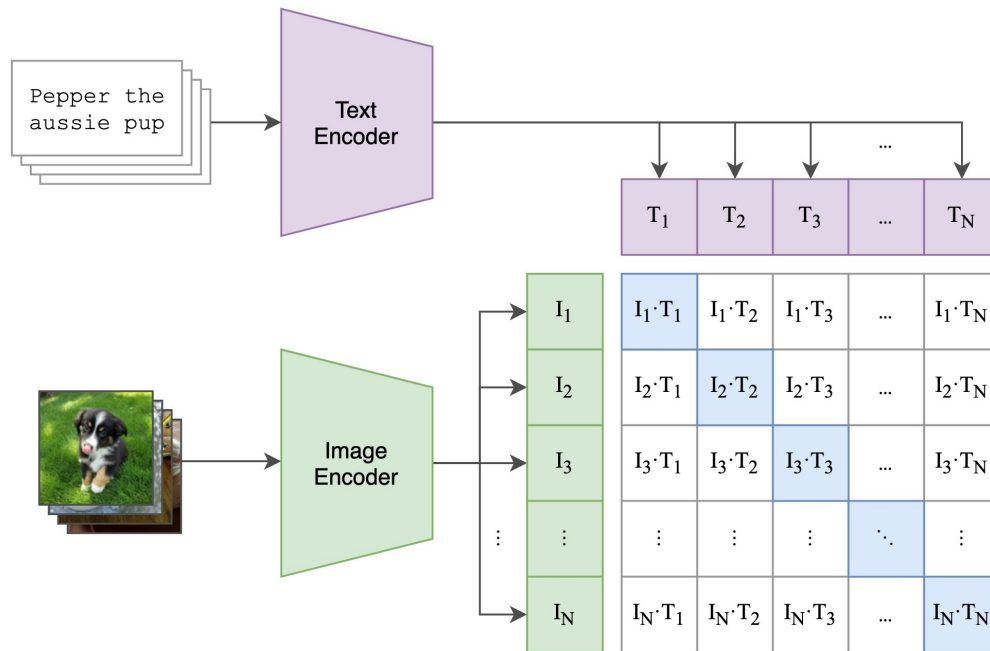
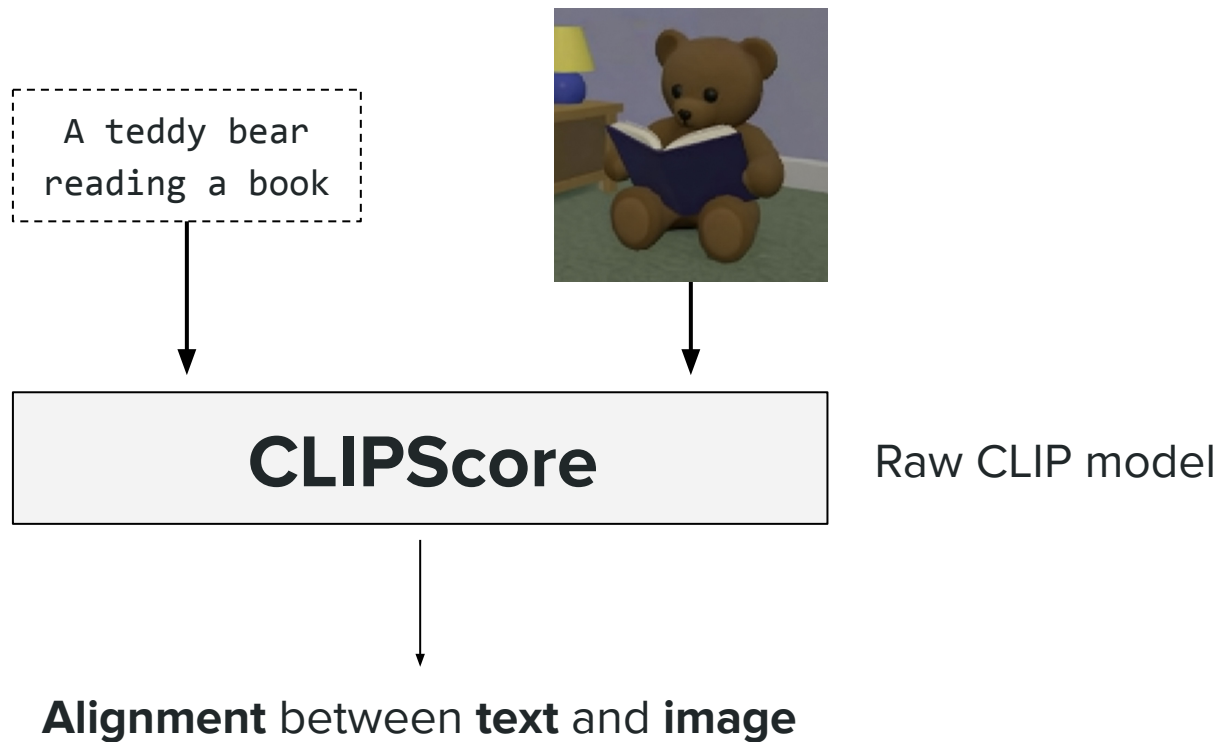
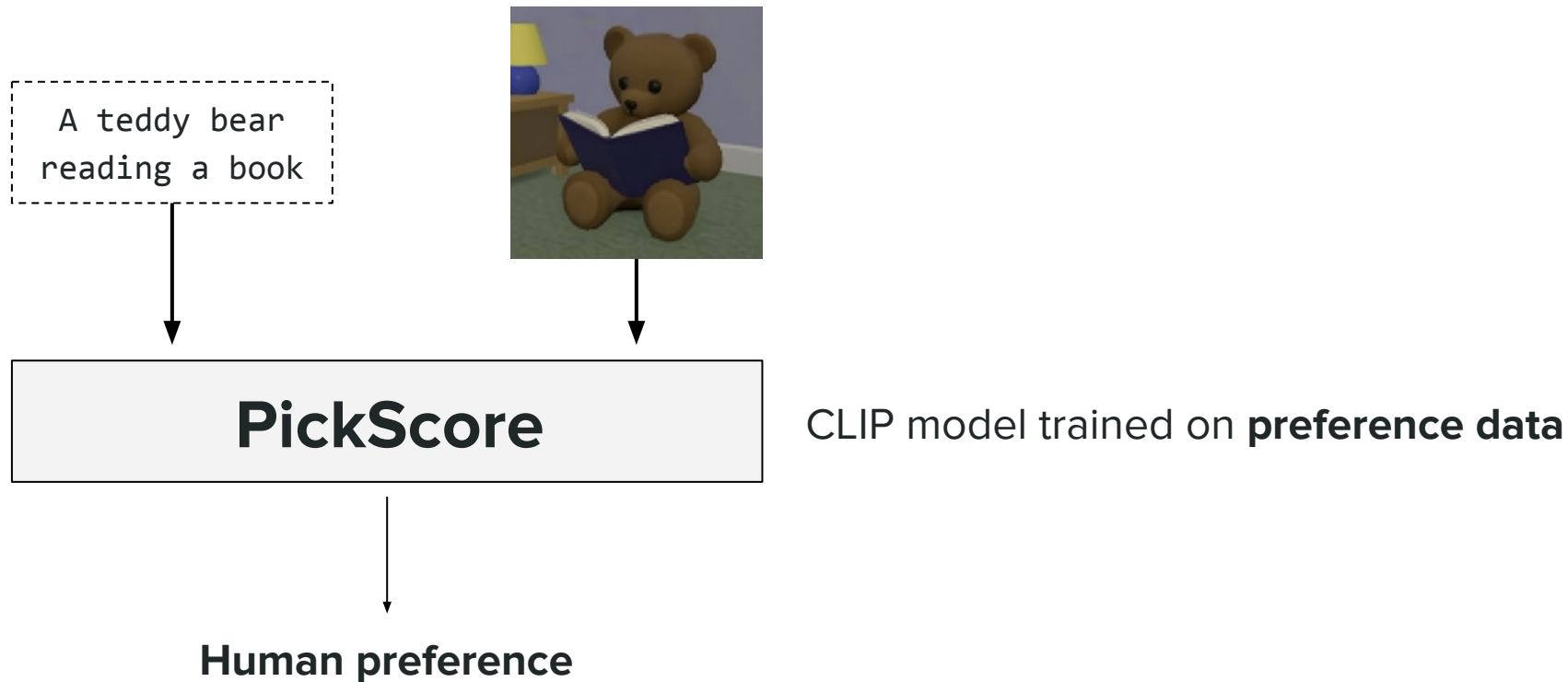


Image-text alignment via CLIPScore



Human preference with PickScore





Diffusion & Large Vision Models

Human ratings

Reference-free metrics

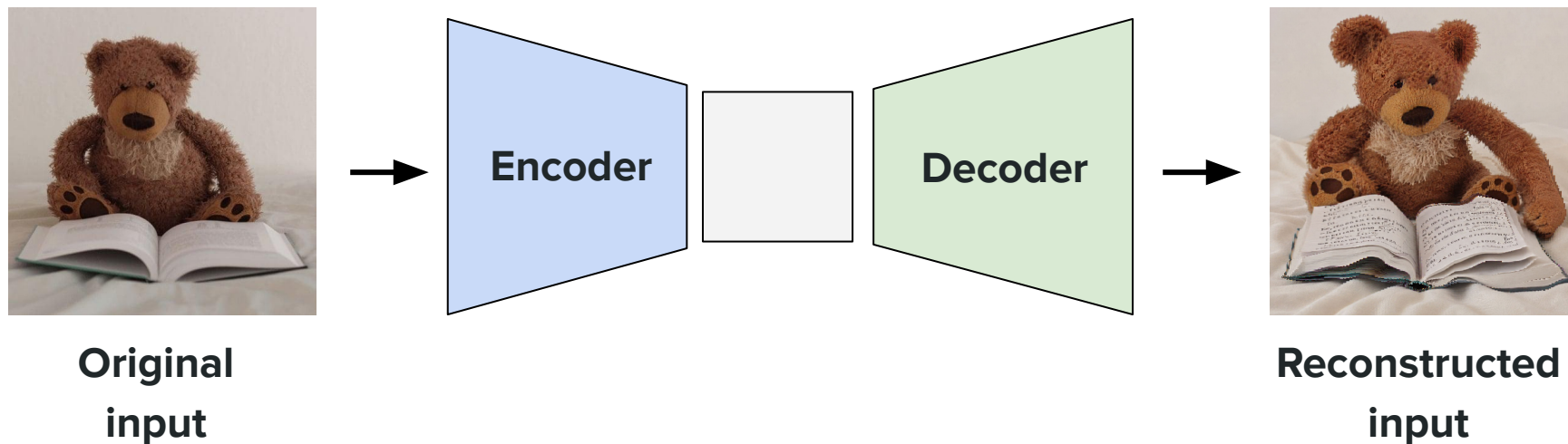
Reference-based metrics

Multimodal LLMs

MLLM-as-a-Judge

Benchmarks

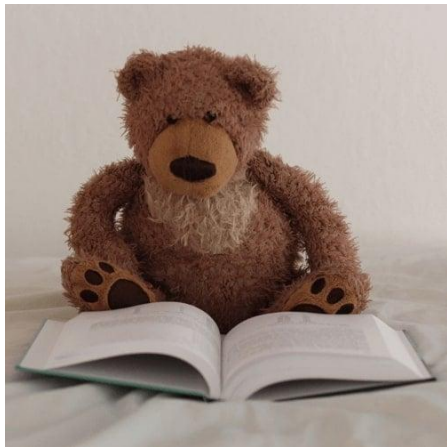
Cases where reference image is available



Other tasks where reference is available. Image editing, distillation

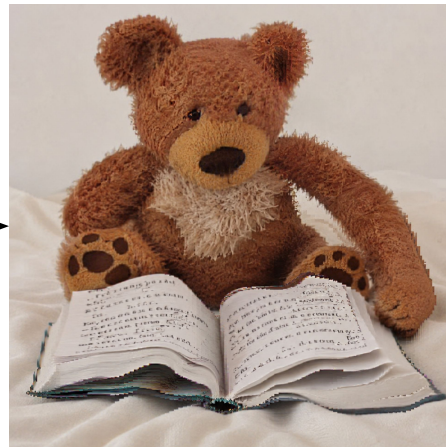
Notations

x



**Original
input**

?



**Reconstructed
input**

$\hat{x}$

Pixel-wise comparisons

MSE = Mean Squared Error

Idea. Pixel-wise distance

$$\text{MSE}(x, \hat{x}) = \frac{1}{HWC} \sum_{i=1}^H \sum_{j=1}^W \sum_{c=1}^C (x_{ijc} - \hat{x}_{ijc})^2$$

Discussion. Not interpretable and sensitive to pixel position

Pixel-wise comparisons

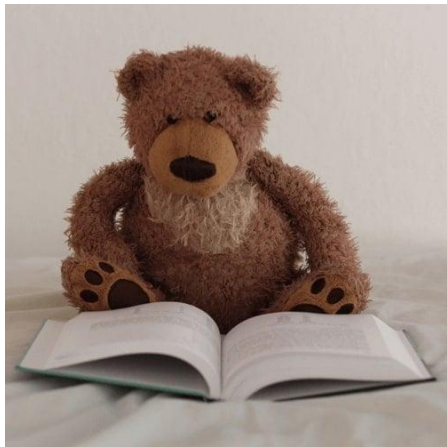
PSNR = **P**eak **S**ignal-to-**N**oise **R**atio

Idea. Normalized pixel-wise distance

$$\text{PSNR}(x, \hat{x}) = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}(x, \hat{x})} \right)$$

Discussion. More interpretable but still sensitive to pixel position

Structure comparison

 x 

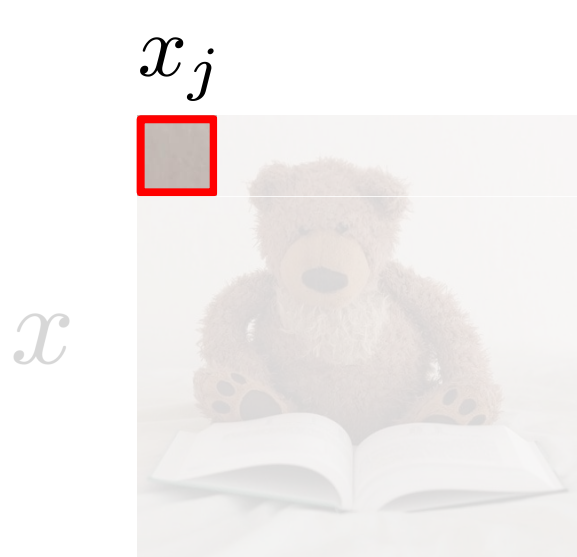
**Original
input**



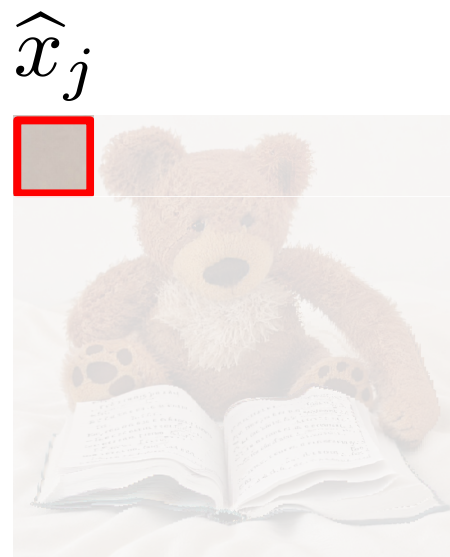
**Reconstructed
input**

 $\hat{x}$

Structure comparison



Original
input



Reconstructed
input

Structure comparison

- **Luminance.** Same brightness? $\longrightarrow$ **Mean** μ
- **Contrast.** Same variance? $\longrightarrow$ **Variance** σ^2
- **Structure.** Correlation between pixels? $\longrightarrow$ **Covariance** $\sigma_{x\hat{x}}^2$

Structure comparison

- **Luminance** similarity $\frac{2\mu_{x_j}\mu_{\hat{x}_j} + c_1}{\mu_{x_j}^2 + \mu_{\hat{x}_j}^2 + c_1}$

Structure comparison

- **Luminance** similarity

$$\frac{2\mu_{x_j} \mu_{\hat{x}_j} + c_1}{\mu_{x_j}^2 + \mu_{\hat{x}_j}^2 + c_1}$$

- **Contrast** similarity

$$\frac{2\sigma_{x_j} \sigma_{\hat{x}_j} + c_2}{\sigma_{x_j}^2 + \sigma_{\hat{x}_j}^2 + c_2}$$

Structure comparison

- **Luminance** similarity $\frac{2\mu_{x_j}\mu_{\hat{x}_j} + c_1}{\mu_{x_j}^2 + \mu_{\hat{x}_j}^2 + c_1}$

- **Contrast** similarity $\frac{2\sigma_{x_j}\sigma_{\hat{x}_j} + c_2}{\sigma_{x_j}^2 + \sigma_{\hat{x}_j}^2 + c_2}$

- **Structure** similarity $\frac{\sigma_{x_j}\hat{x}_j + c_3}{\sigma_{x_j}\sigma_{\hat{x}_j} + c_3}$

Structure comparison

- **Luminance** similarity $\frac{2\mu_{x_j}\mu_{\hat{x}_j} + c_1}{\mu_{x_j}^2 + \mu_{\hat{x}_j}^2 + c_1} \quad l(x_j, \hat{x}_j)$
- **Contrast** similarity $\frac{2\sigma_{x_j}\sigma_{\hat{x}_j} + c_2}{\sigma_{x_j}^2 + \sigma_{\hat{x}_j}^2 + c_2} \quad c(x_j, \hat{x}_j)$
- **Structure** similarity $\frac{\sigma_{x_j\hat{x}_j} + c_3}{\sigma_{x_j}\sigma_{\hat{x}_j} + c_3} \quad s(x_j, \hat{x}_j)$

Structure comparison

$$\text{SSIM}(x_j, \hat{x}_j) = \underbrace{[l(x_j, \hat{x}_j)]}_{\text{Luminance similarity}} \cdot \underbrace{[c(x_j, \hat{x}_j)]}_{\text{Contrast similarity}} \cdot \underbrace{[s(x_j, \hat{x}_j)]}_{\text{Structure similarity}}$$

Structure comparison

$$\text{SSIM}(x_j, \hat{x}_j) = \frac{(2\mu_{x_j} \mu_{\hat{x}_j} + c_1)(2\sigma_{x_j \hat{x}_j} + c_2)}{(\mu_{x_j}^2 + \mu_{\hat{x}_j}^2 + c_1)(\sigma_{x_j}^2 + \sigma_{\hat{x}_j}^2 + c_2)}$$

Structure comparison



Structure comparison

SSIM = Structural **SIM**ilarity

Idea. Measure structural similarity by comparing patch statistics

$$\text{SSIM}(x, \hat{x}) = \frac{1}{M} \sum_{j=1}^M \text{SSIM}(x_j, \hat{x}_j)$$

Limitations. Still vulnerable to "spatial shift"

Feature comparison

LPIPS = **L**earned **P**erceptual **I**mage **P**atch **S**imilarity

Idea. Use pre-trained model to **compute features** that align with **visual perception**

$$\text{LPIPS}(x, \hat{x}) = \sum_l \frac{1}{H_l W_l} \left\| w^l \odot (\phi_l(x) - \phi_l(\hat{x})) \right\|^2$$

Limitations. Not directly interpretable



Diffusion & Large Vision Models

Human ratings

Reference-free metrics

Reference-based metrics

Multimodal LLMs

MLLM-as-a-Judge

Benchmarks

Objective

Gap. Fixed metrics are not interpretable.

- Many of them
- Operate at different levels
- Fragmentation of meaning
- Inevitable misalignments

Objective

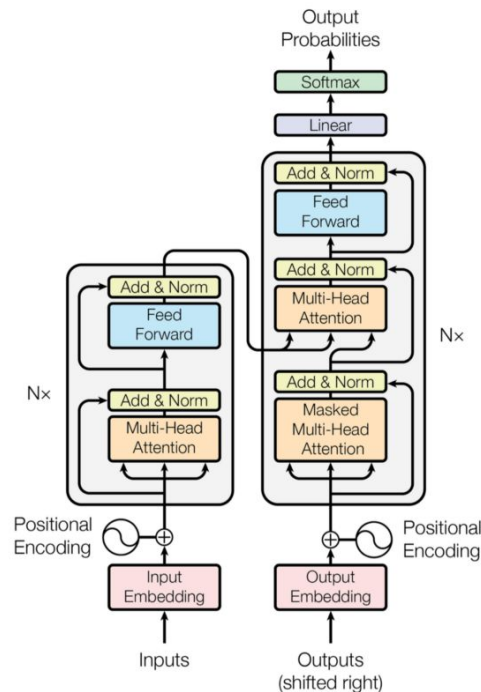
Gap. Fixed metrics are not interpretable.

- Many of them
- Operate at different levels
- Fragmentation of meaning
- Inevitable misalignments

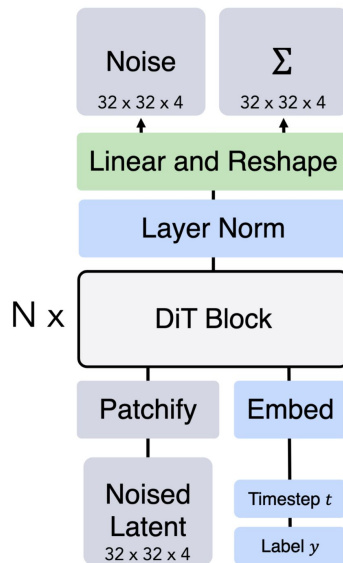
Detour. Model that "speaks"? LLMs!

Any way to adapt wins to images?

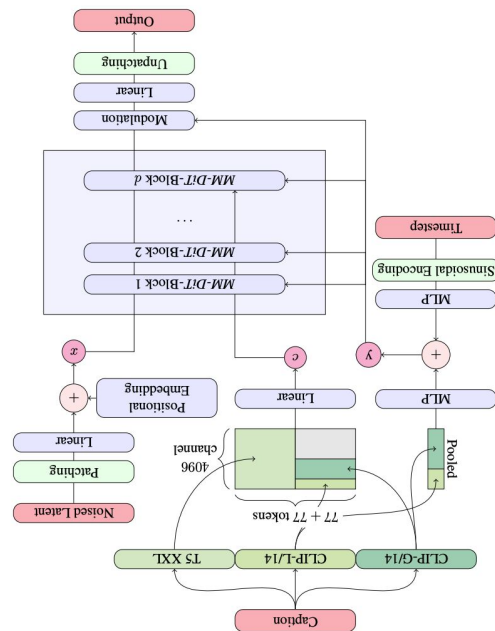
Objective



Transformer

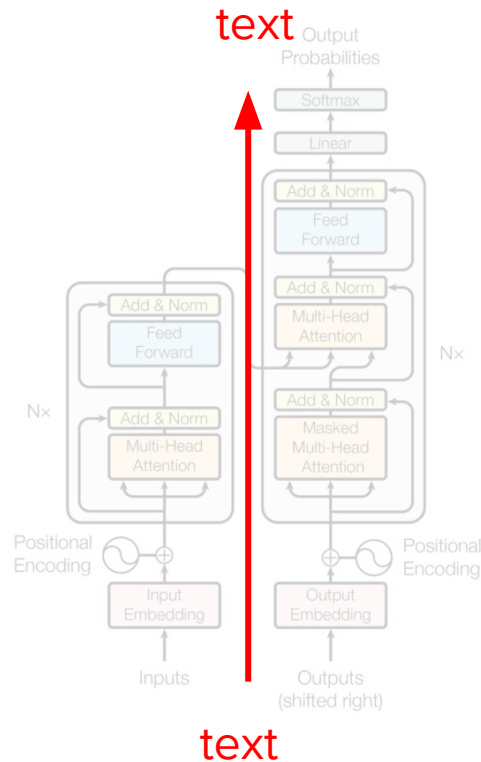


DiT

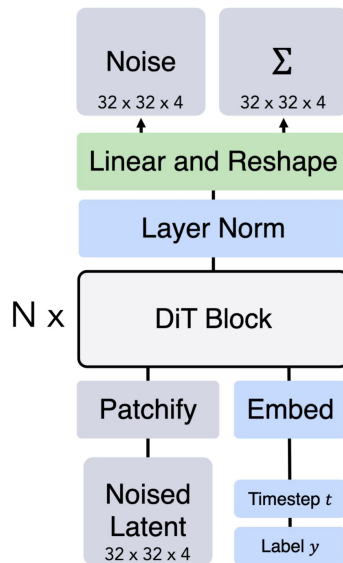


MM-DiT

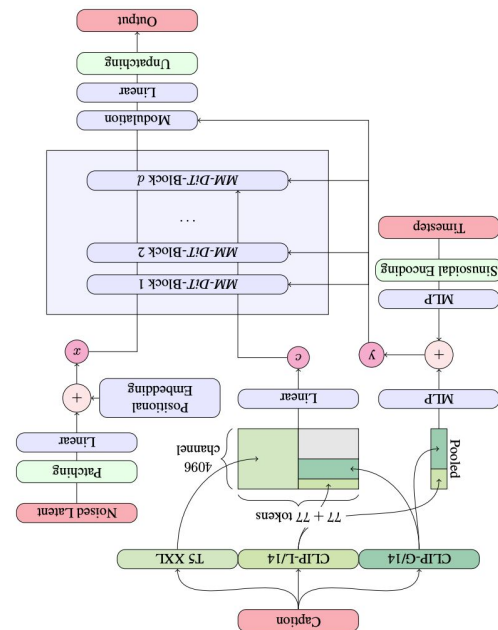
Objective



Transformer

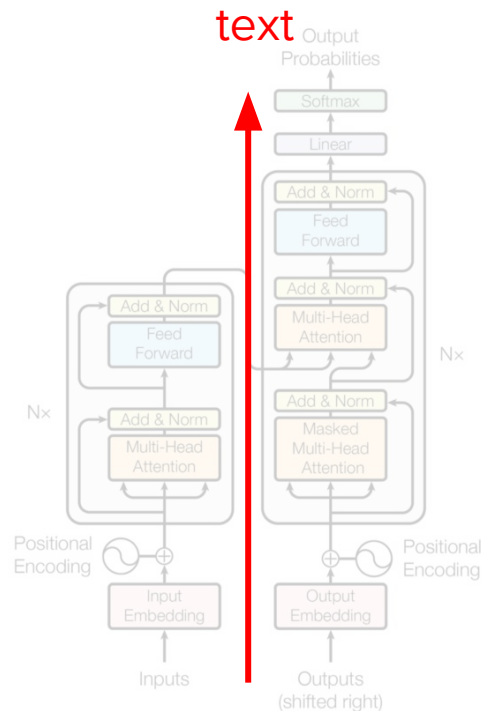


DiT

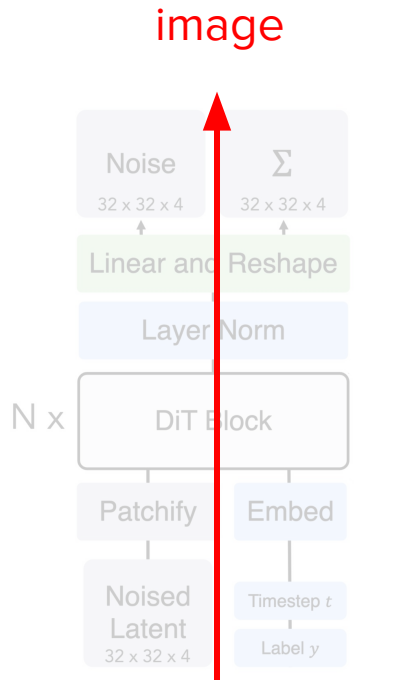


MM-DiT

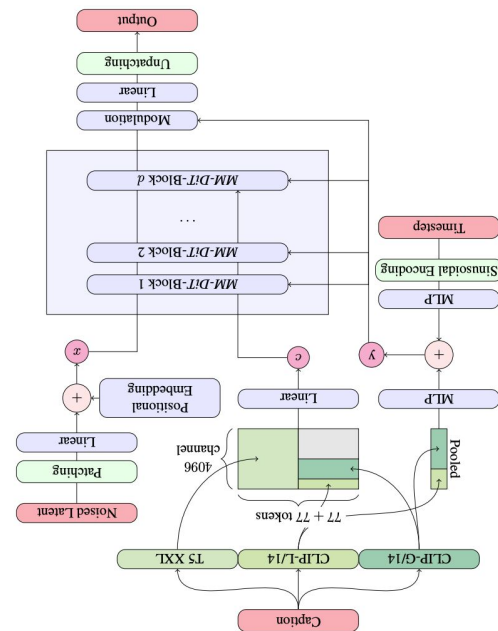
Objective



Transformer

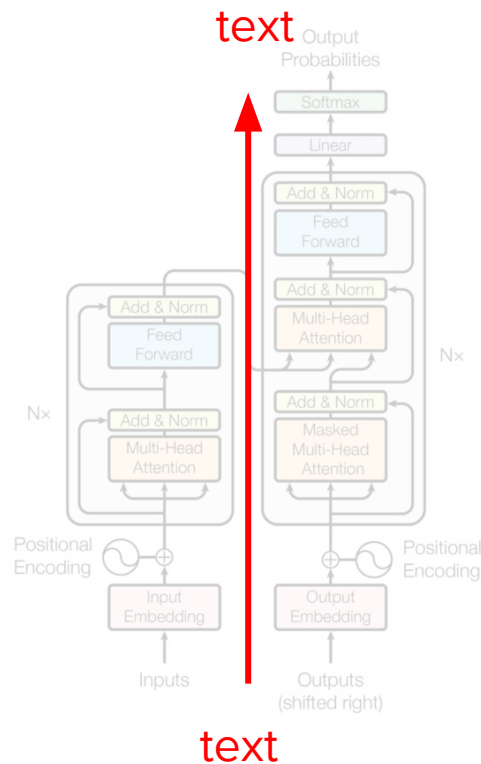


DiT

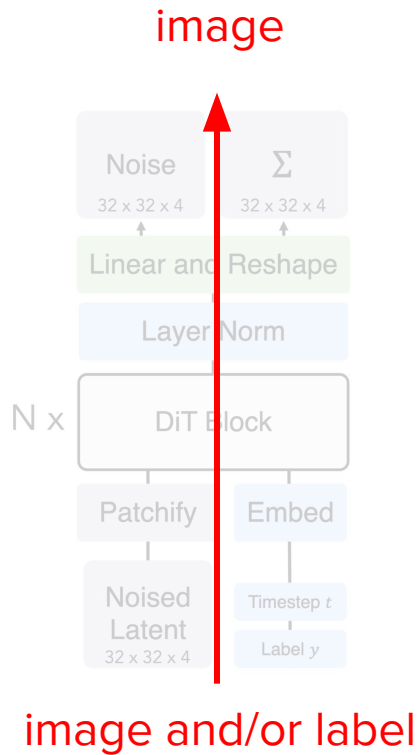


MM-DiT

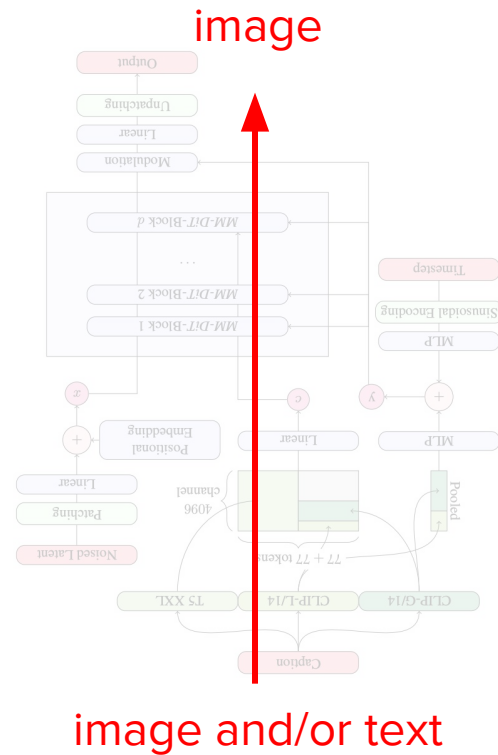
Objective



Transformer



DiT



MM-DiT

Objective

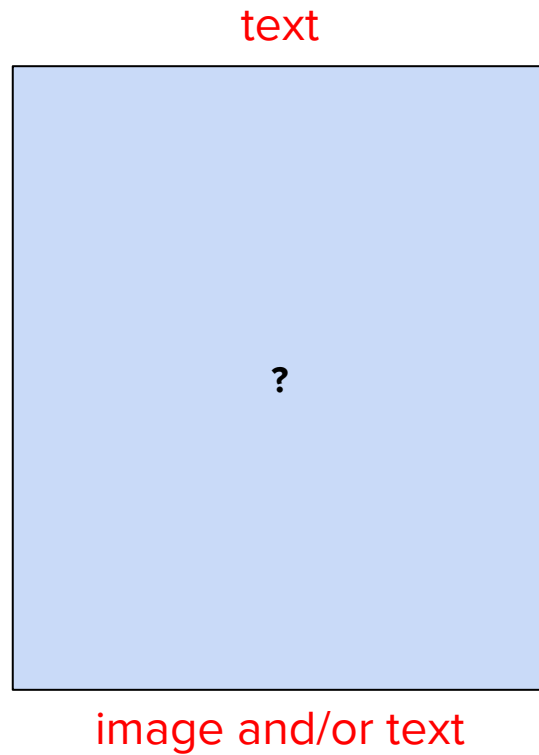
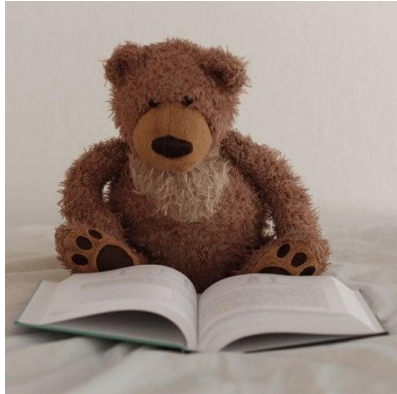


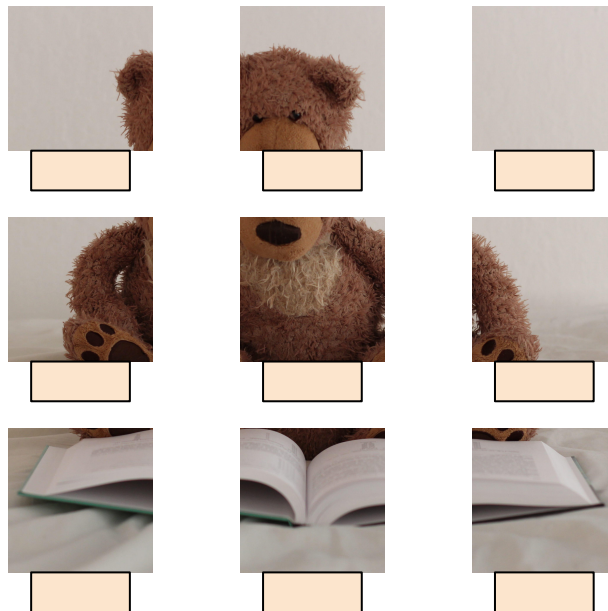
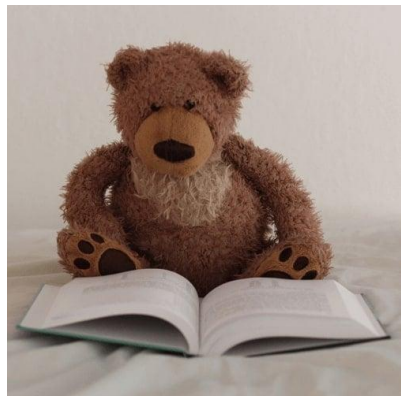
Image understanding based on text input



Very cute!

How cute is this teddy bear? →

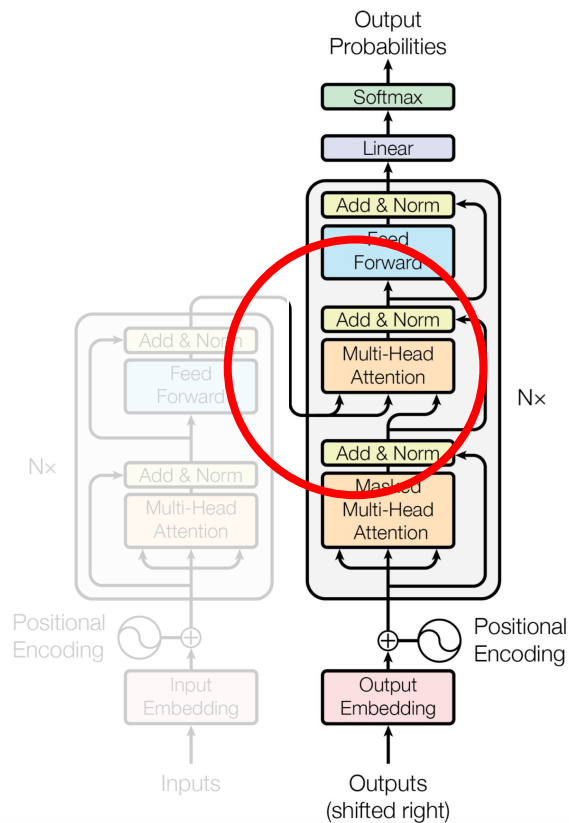
Tokenization



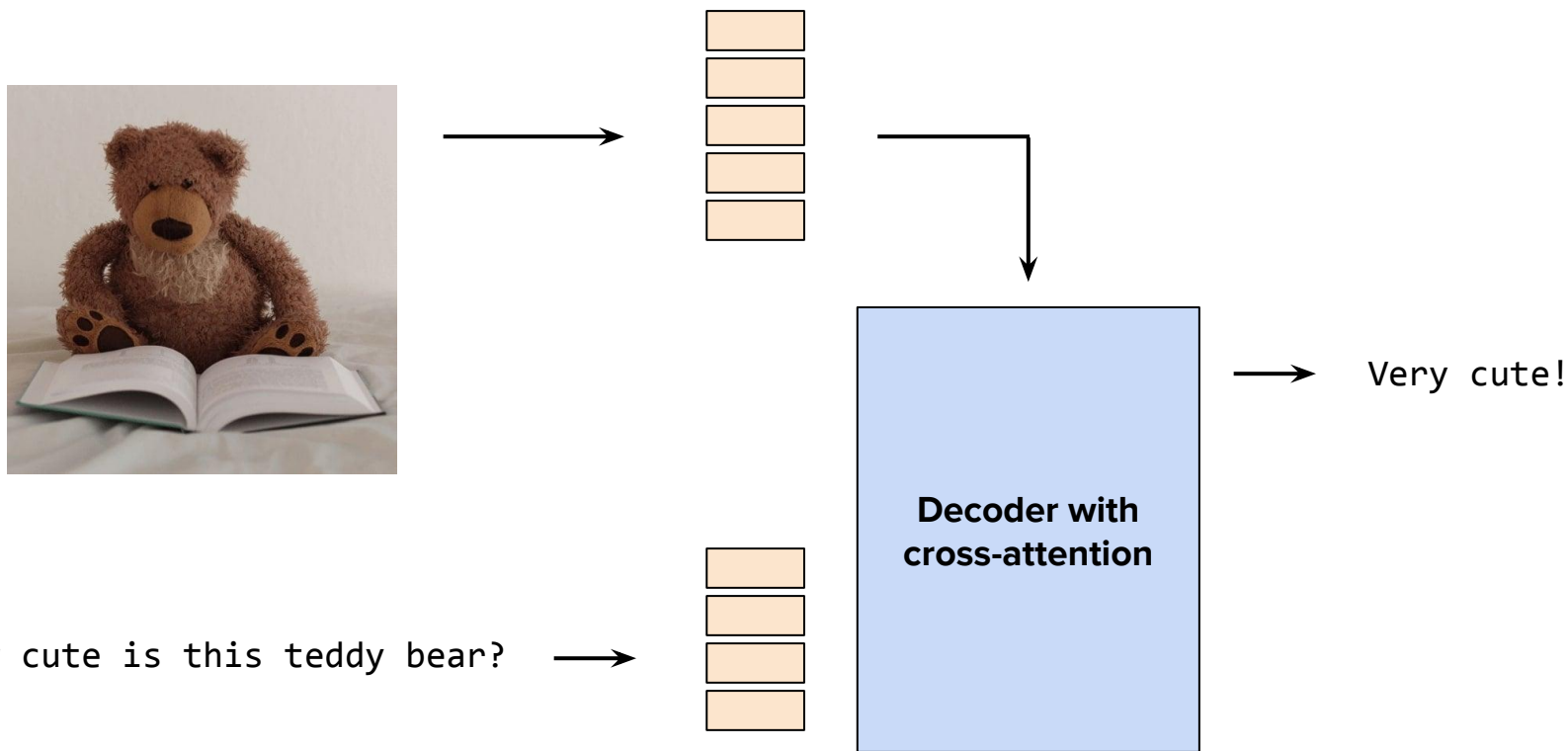
How cute is this teddy bear?



Idea 1: leverage the original Transformer!

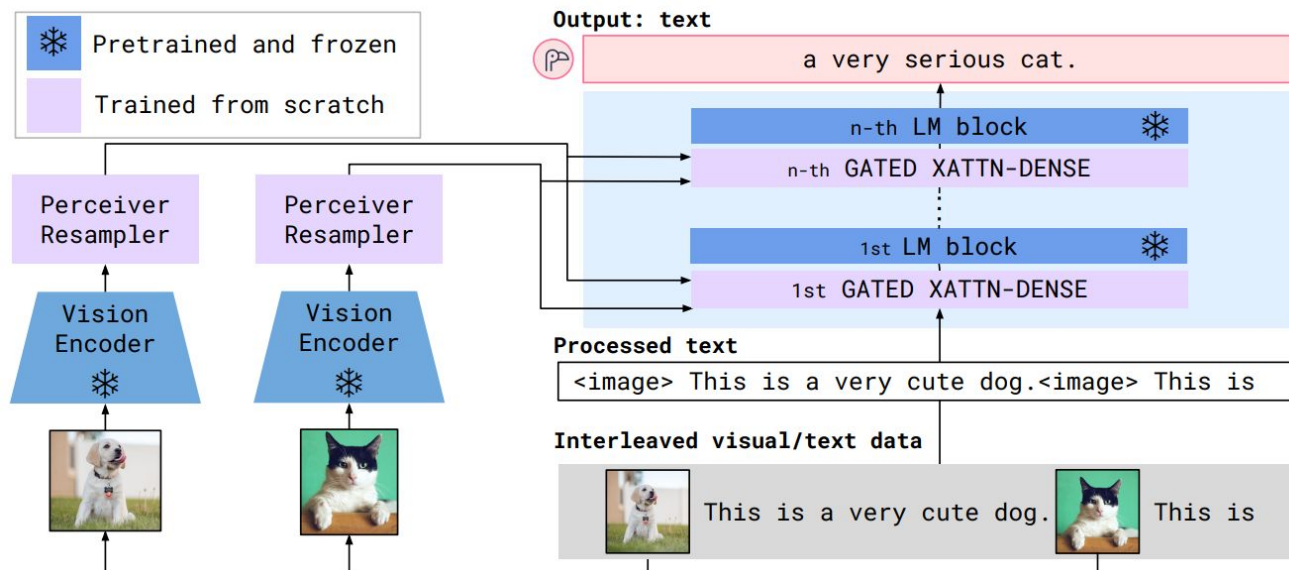


Idea 1: leverage the original Transformer!



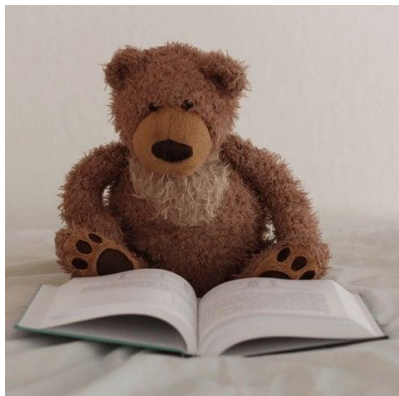
Idea 1: leverage the original Transformer!

Idea. Encode image modality as cross-attention keys/values.



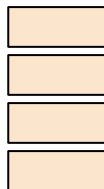
Flamingo (2022)

Idea 2: recycle decoder-only architecture



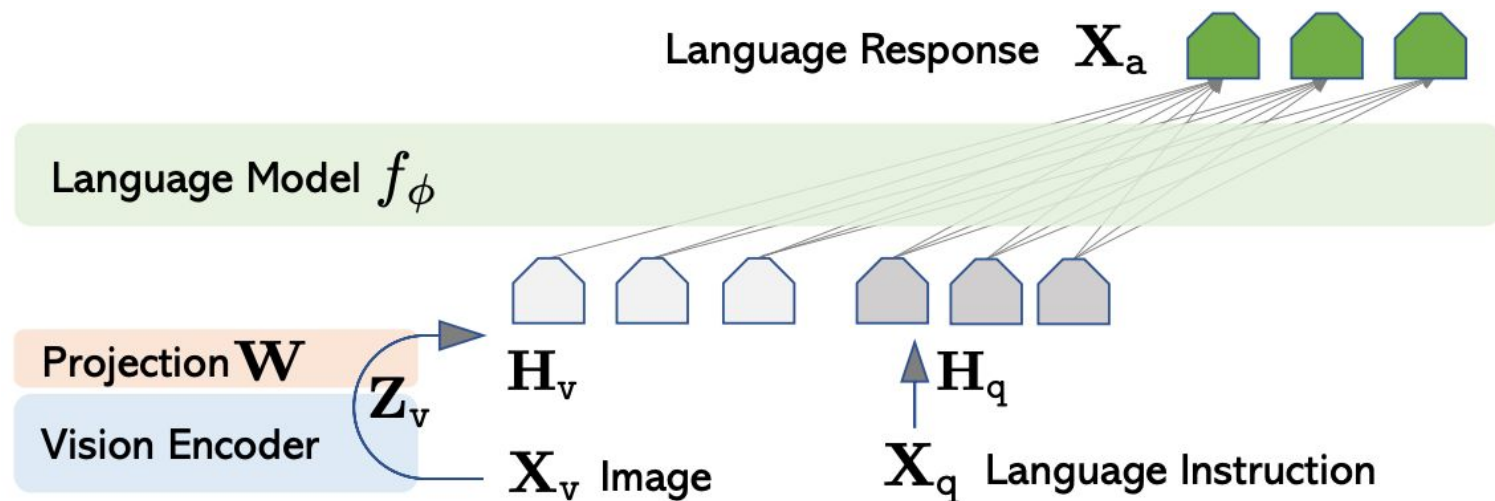
Very cute!

How cute is this teddy bear?



Idea 2: recycle decoder-only architecture

Idea. Just treat it as a regular LLM input!



LLaVA (2023)

Architecture. Decoder with joint attention.

Trends

Architecture. Decoder with joint attention.

Capabilities.

- Work across resolutions
- Spatial awareness
- OCR
- Includes other modalities
- Reasoning



Diffusion & Large Vision Models

Human ratings

Reference-free metrics

Reference-based metrics

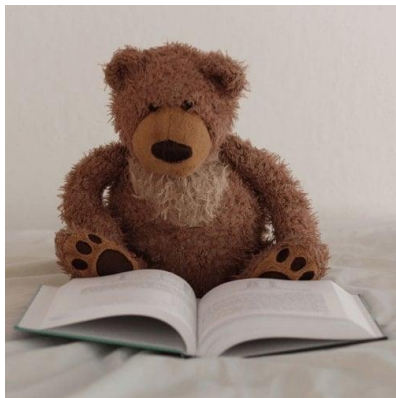
Multimodal LLMs

MLLM-as-a-Judge

Benchmarks

Motivation

Complaint. Traditional metrics are both holistic and opaque.



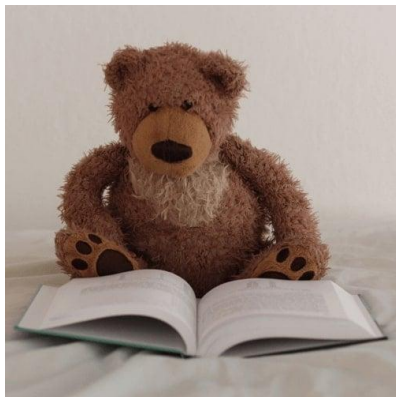
A cute teddy bear is reading a book

CLIPScore: 0.922

why?

TIFA = Text-to-Image Faithfulness Evaluation with QA

Idea. Decompose faithfulness into atomic questions.



Is there a teddy bear?



Is the teddy bear cute?



Is there a book?



Is the teddy bear reading the book?



A cute teddy bear is reading a book

TIFA: 100.0

Benefits.

- Simple to understand
- Score debugging is easy

Benefits.

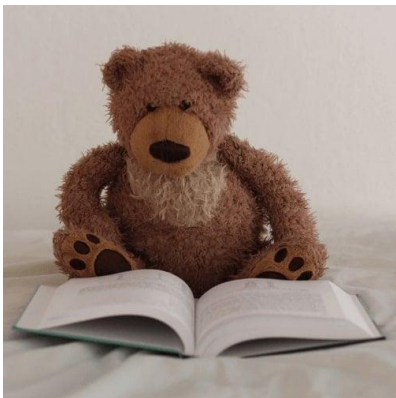
- Simple to understand
- Score debugging is easy

Drawbacks.

- Not easy to convey per-claim importance
- Question generation itself error-prone

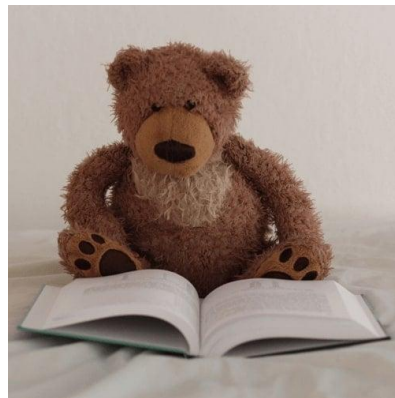
Motivation

Complaint. Traditional metrics do not carry composition well.



A cute **teddy bear** is reading a **book**

CLIPScore

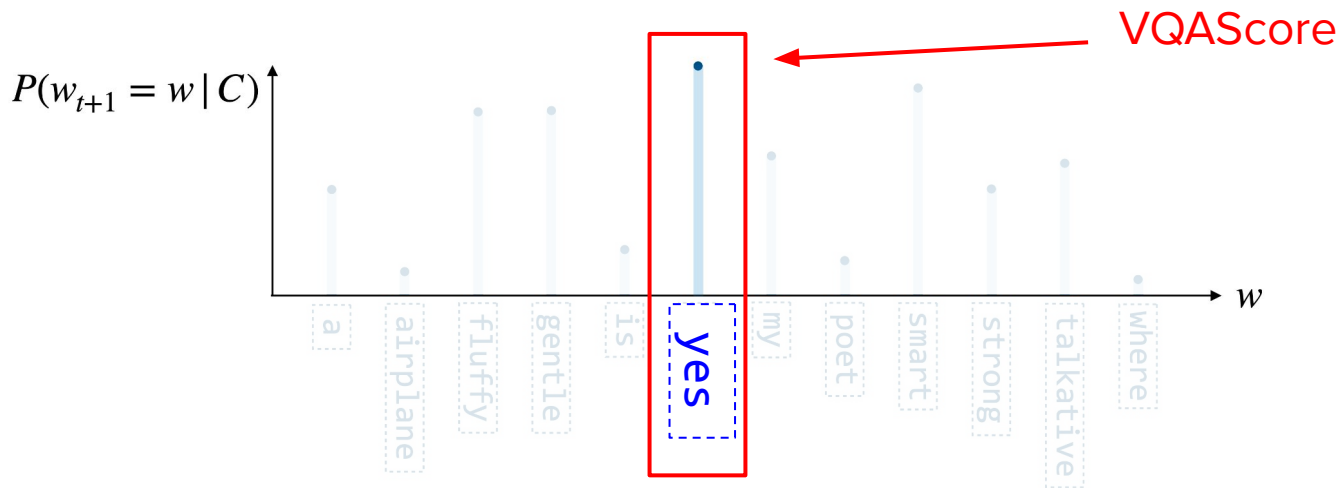


A cute **book** is reading a **teddy bear**

CLIPScore



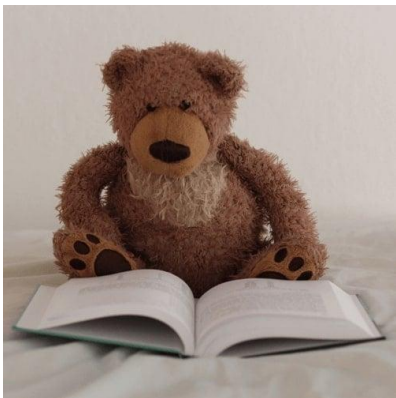
VQAScore = Visual **Q**uestion **A**nswering **S**core



[Image] Does this figure show [Prompt]? Please answer yes or no.

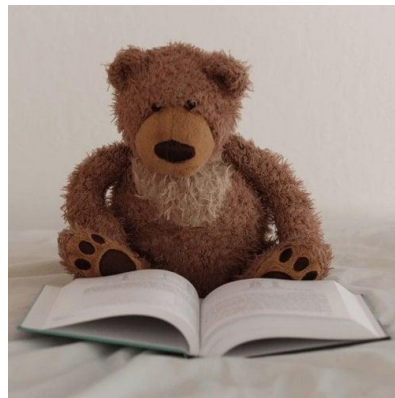
VQAScore

Benefit. Leverages LLM-style understanding capabilities.



A cute **teddy bear** is reading a **book**

VQAScore



A cute **book** is reading a **teddy bear**

VQAScore

Limitations.

- Assumes access to token probabilities
- TIFA + VQAScore means multiple MLLM calls
- Feels like there should be a more generic approach

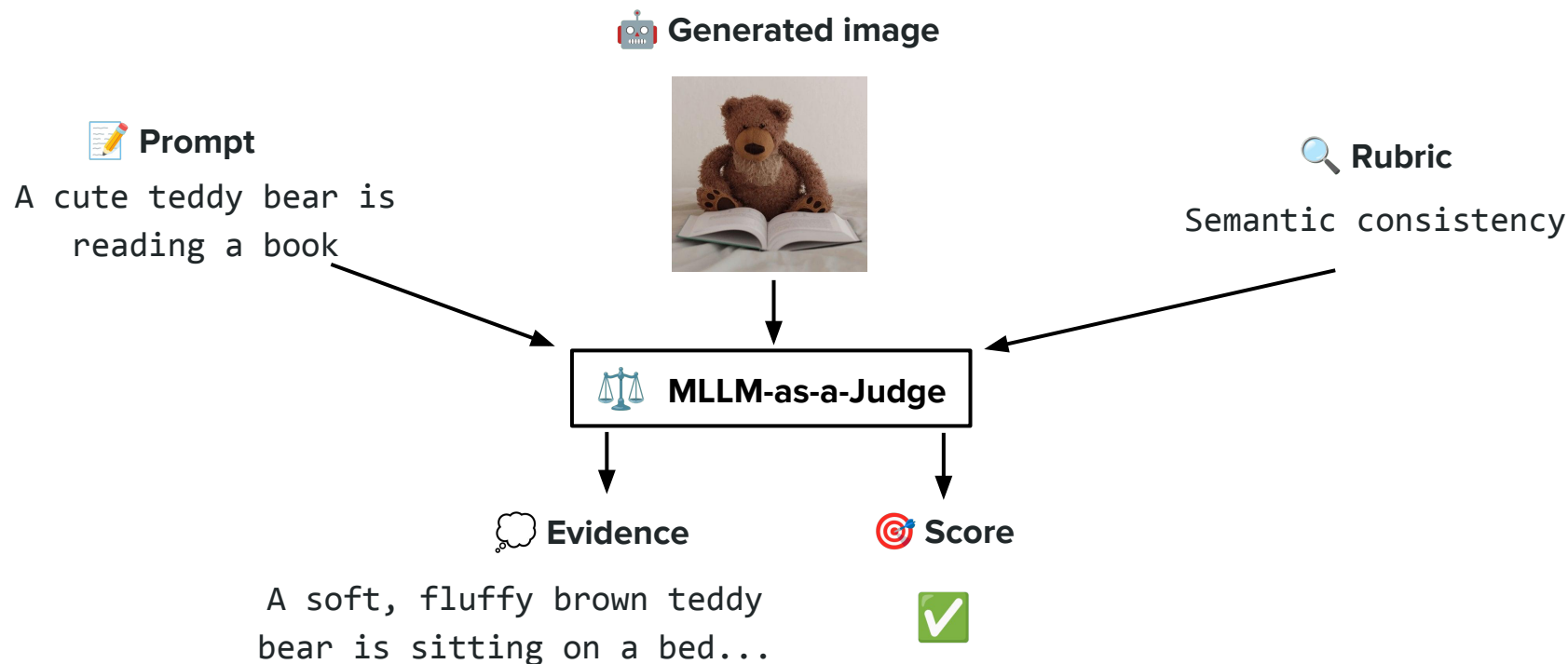
Limitations.

- Assumes access to token probabilities
- TIFA + VQAScore means multiple MLLM calls
- Feels like there should be a more generic approach

Attractive alternative mindset

Engineer outcome  **Infer** outcome

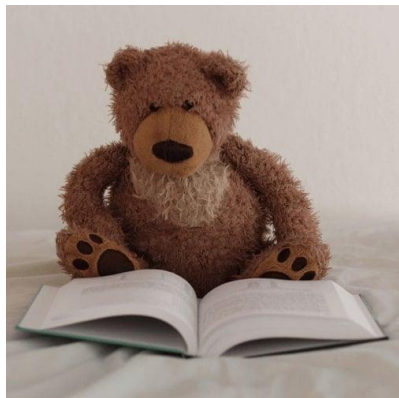
VIEScore = Visual Instruction-guided Explainable Score



Dimensions

Example. Rubrics could be:

- **Semantic consistency:** How aligned with the conditions?
- **Perceptual quality:** How authentic and natural?



Structure

Example of a judge prompt.

Input: A cute teddy bear is reading a book,

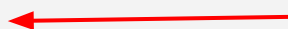


Rubric: Please the input across the following axes

- Dimension 1: Guidelines...
- Dimension 2: Guidelines...

Output: Return the graded result in JSON format:

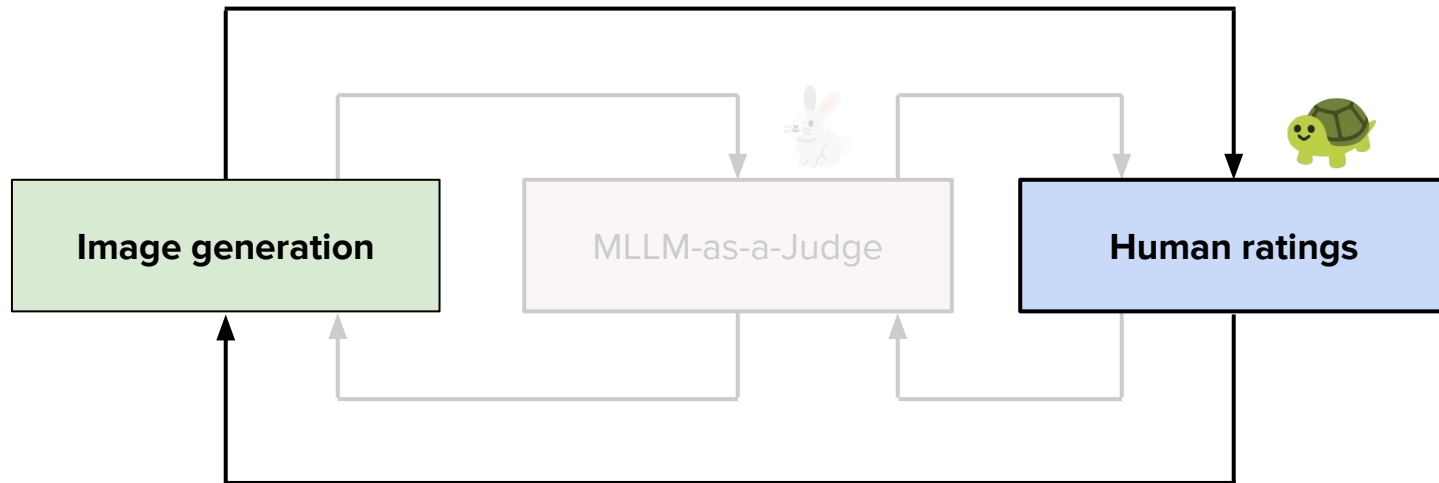
```
{  
  "rationale": ...,  
  "score": ...,  
}
```



How to ensure calibration?

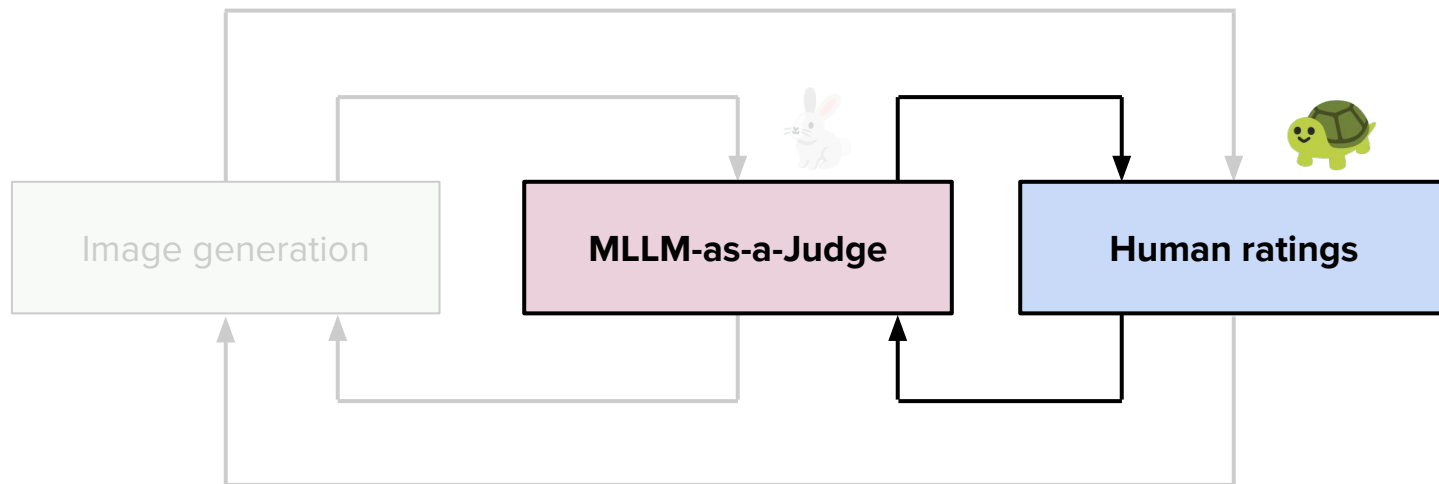
Typical workflow

Seed. Seek human expertise about task at hand.



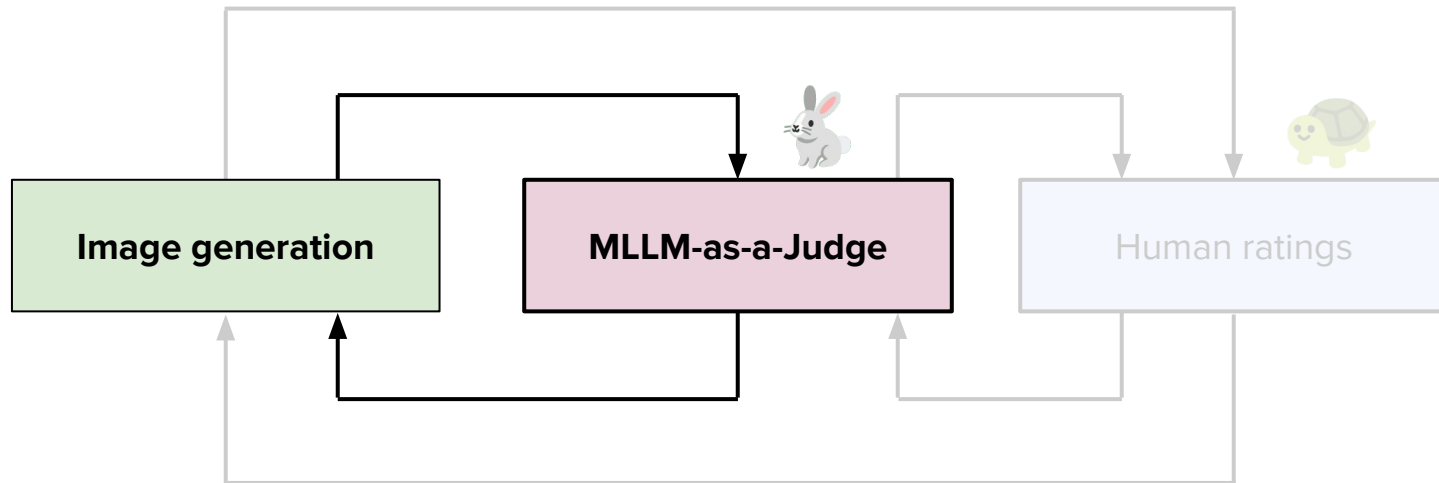
Typical workflow

Calibrate. Align MLLM-as-a-Judge with human expertise.



Typical workflow

Automate. Rely on MLLM-as-a-Judge for general cases.



Use cases

Pointwise

Evaluate the quality
of:

Response



Very good

Pairwise

Which one is better:

Response A

or

Response B

?



Response A

Batch ranking

Rank the following:

Response A

Response B

Response C

...



B → A → C

Use cases

Pointwise

Evaluate the quality
of:

Response



Very good

great for diagnostics

Pairwise

Which one is better:

Response A

or

Response B

?



Response A

Batch ranking

Rank the following:

Response A

Response B

Response C

...



2/1/3

Use cases

Pointwise

Evaluate the quality
of:

Response



Very good

great for diagnostics

Pairwise

Which one is better:

Response A

or

Response B

?



Response A

great to compare models

Batch ranking

Rank the following:

Response A

Response B

Response C

...



B → A → C

Use cases

Pointwise

Evaluate the quality
of:

Response



Very good

great for diagnostics

Pairwise

Which one is better:

Response A
or
Response B
?



Response A

great to compare models

Batch ranking

Rank the following:

Response A
Response B
Response C
...



B → A → C

sensitive to ordering
not really used in practice

Best practices

- Parse score into **atomic criteria**

Best practices

- Parse score into **atomic criteria**
- Ask MLLM to describe **evidence** before giving a score

Best practices

- Parse score into **atomic criteria**
- Ask MLLM to describe **evidence** before giving a score
- Use **low temperature** and structured outputs

Best practices

- Parse score into **atomic criteria**
- Ask MLLM to describe **evidence** before giving a score
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- **Randomize** A/B order for pairwise judging

Best practices

- Parse score into **atomic criteria**
- Ask MLLM to describe **evidence** before giving a score
- Use **low temperature** and structured outputs
- **Randomize** A/B order for pairwise judging
- Validate judge scores against **human ratings** before trusting it



Diffusion & Large Vision Models

Human ratings

Reference-free metrics

Reference-based metrics

Multimodal LLMs

MLLM-as-a-Judge

Benchmarks

Common benchmarks

Object alignment

- Verify objects, counts, colors
- Tests compositional image generation
- Fine-grained failure diagnosis

GenEval

Text-to-image alignment example: GenEval

GenEval[?] = **Generation Evaluation**

Goal. Tests rendering specific objects and their attributes.

Characteristics:

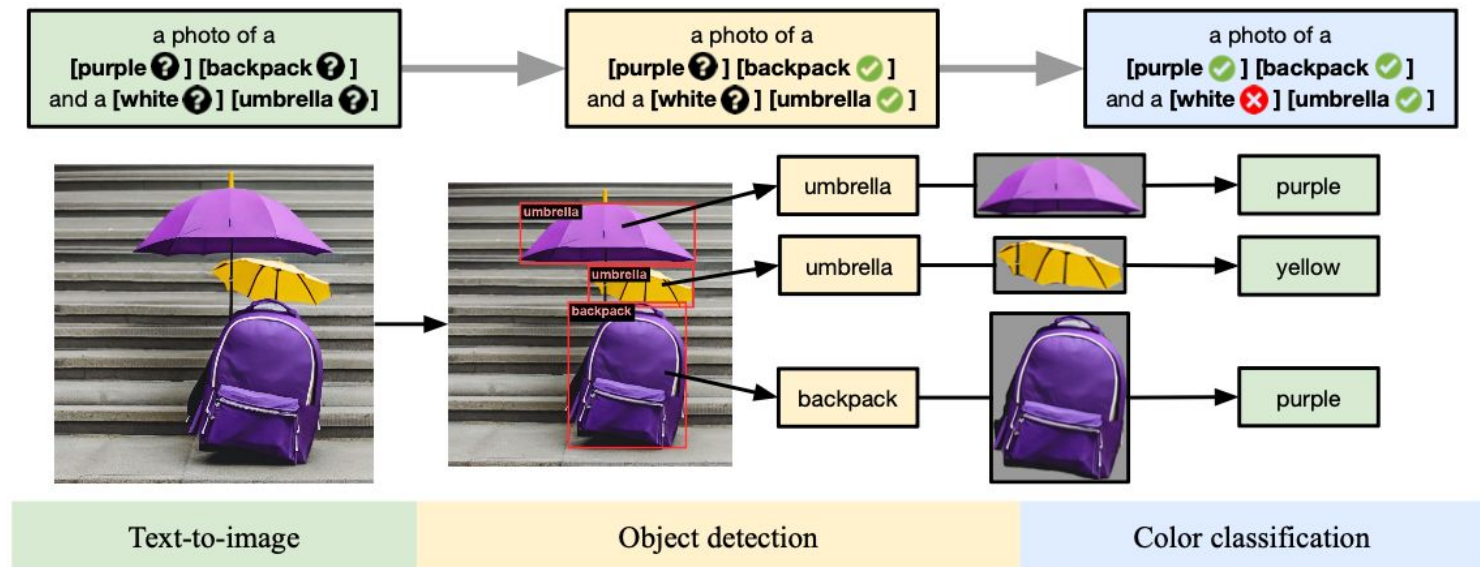
- ~600 prompts across 6 tasks
- Tasks: 1/2 objects, counts, colors, position, color attribution

Object detection,
geometry,
color classification

Evaluation criteria. Yes/no per task associated to prompt.

Text-to-image alignment example: GenEval

Samples.



Common benchmarks

Object alignment

- Verify objects, counts, colors
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GenEval

Dense prompts

- Follow long detailed prompts
- Track entities, attributes, relations
- Evaluates semantic completeness

DPG-Bench

Dense prompt following example: DPG-Bench

DPG-Bench = Dense **P**rompt **G**raph **B**enchmark

Goal. Remembers every detail when given a massive paragraph of text.

Characteristics:

- Graph of questions derived with the "DSG" pipeline into entity/attribute/relation
- ~14,000 questions across ~1,000 prompts

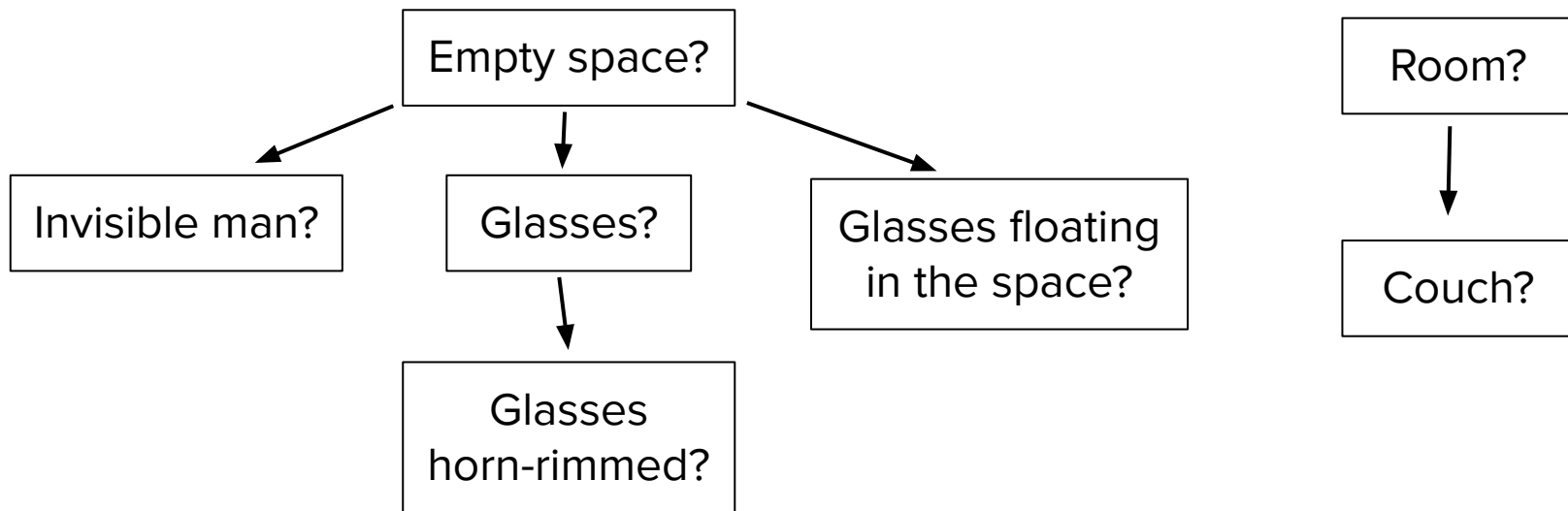
VQA judge in graph

Evaluation criteria. Yes/no to questions in graph.

Dense prompt following example: DPG-Bench

Example.

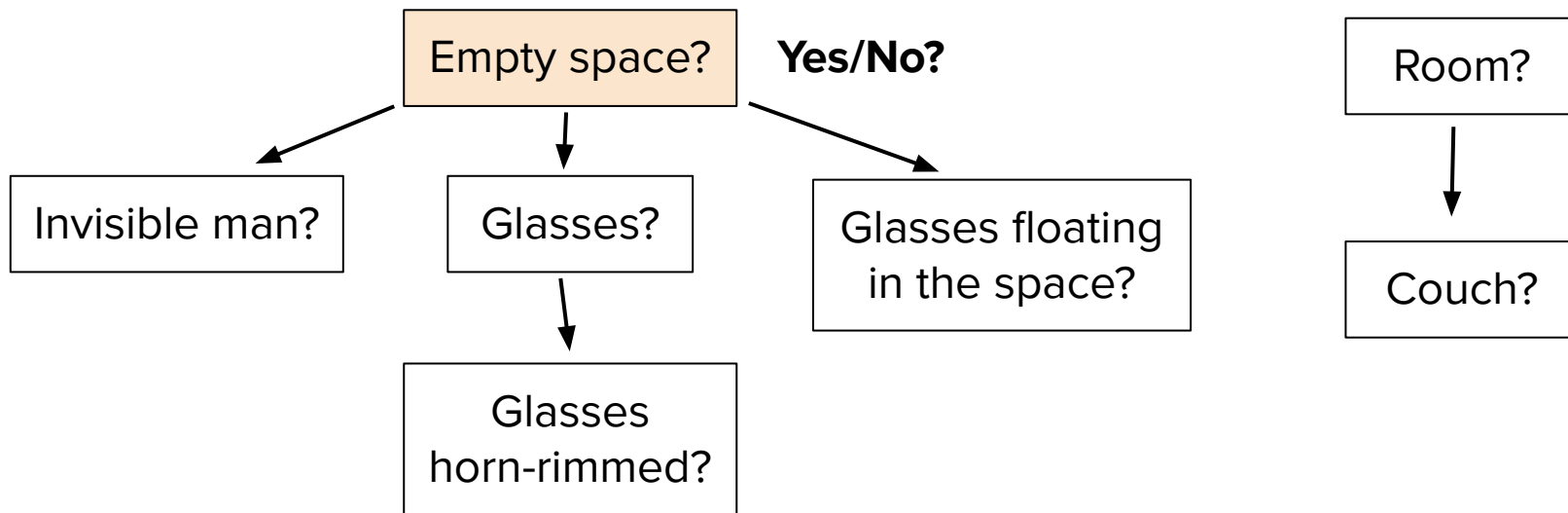
An empty space where an invisible man would be, with a pair of horn-rimmed glasses seemingly floating in mid-air, and a pearl bead necklace draped in the space below them. In the space where his hands would be, a smartphone is held... the room appears ordinary, with a couch and a coffee table nearby, upon which rests a scattering of magazines and a remote control.



Dense prompt following example: DPG-Bench

Example.

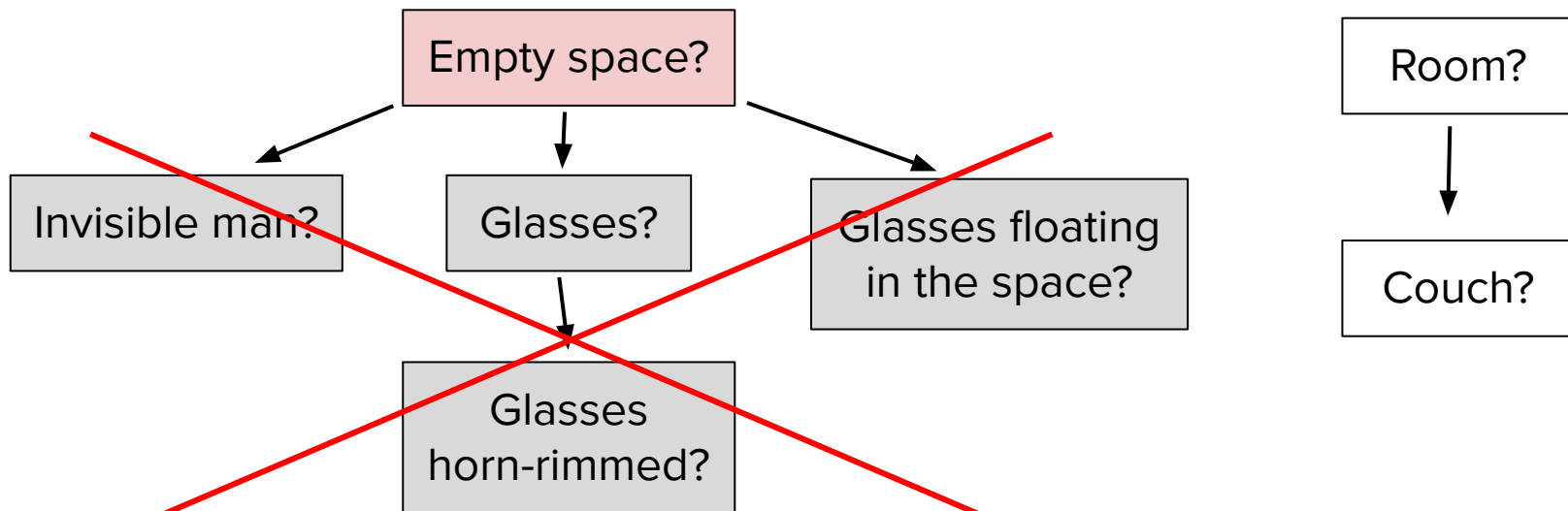
An empty space where an invisible man would be, with a pair of horn-rimmed glasses seemingly floating in mid-air, and a pearl bead necklace draped in the space below them. In the space where his hands would be, a smartphone is held... the room appears ordinary, with a couch and a coffee table nearby, upon which rests a scattering of magazines and a remote control.



Dense prompt following example: DPG-Bench

Example.

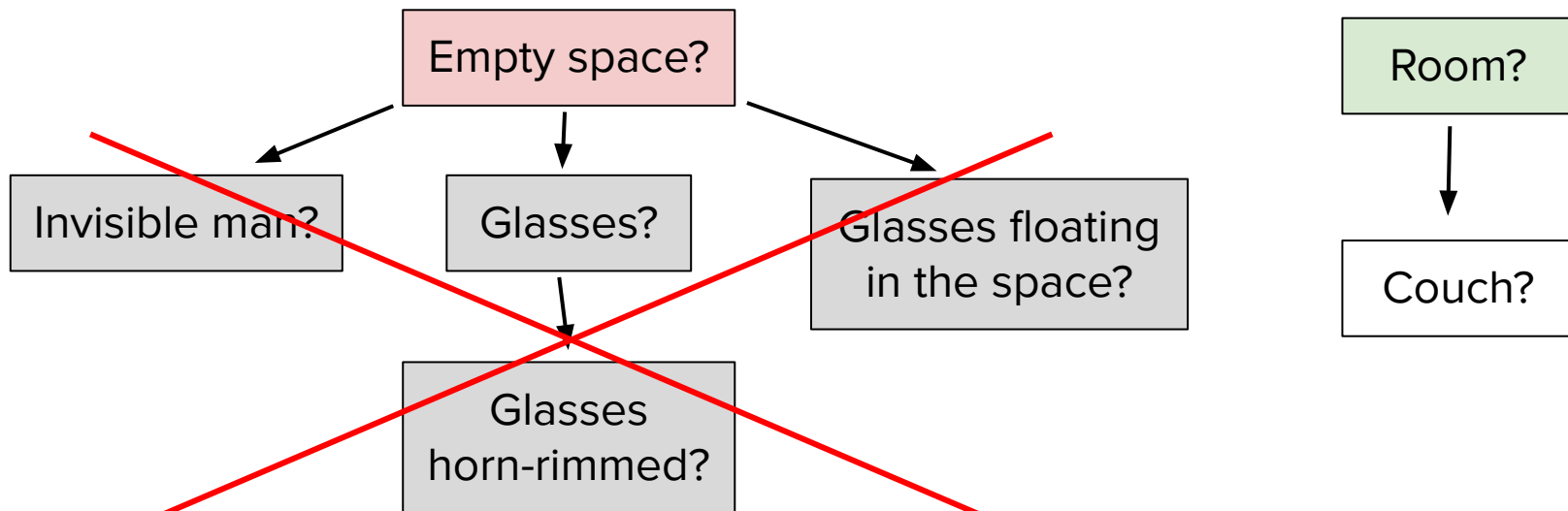
An empty space where an invisible man would be, with a pair of horn-rimmed glasses seemingly floating in mid-air, and a pearl bead necklace draped in the space below them. In the space where his hands would be, a smartphone is held... the room appears ordinary, with a couch and a coffee table nearby, upon which rests a scattering of magazines and a remote control.



Dense prompt following example: DPG-Bench

Example.

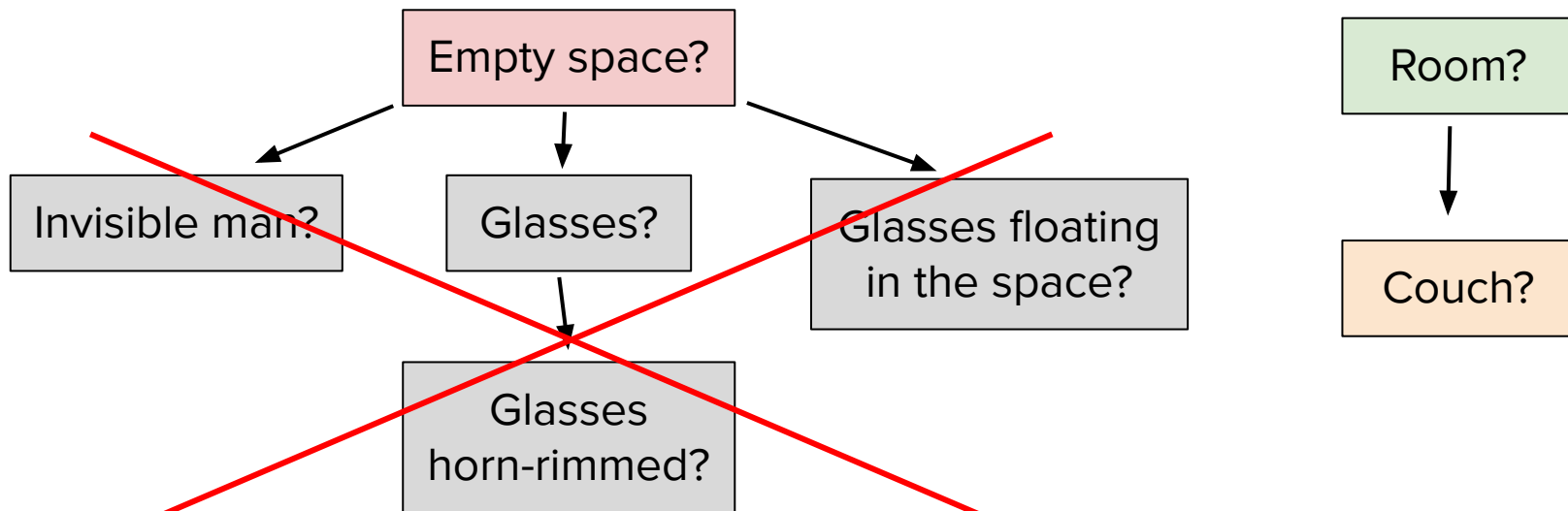
An empty space where an invisible man would be, with a pair of horn-rimmed glasses seemingly floating in mid-air, and a pearl bead necklace draped in the space below them. In the space where his hands would be, a smartphone is held... the room appears ordinary, with a couch and a coffee table nearby, upon which rests a scattering of magazines and a remote control.



Dense prompt following example: DPG-Bench

Example.

An empty space where an invisible man would be, with a pair of horn-rimmed glasses seemingly floating in mid-air, and a pearl bead necklace draped in the space below them. In the space where his hands would be, a smartphone is held... the room appears ordinary, with a couch and a coffee table nearby, upon which rests a scattering of magazines and a remote control.



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DPG-Bench

Text rendering

- Generate readable text in images
- Handles long strings
- Tests OCR-level fidelity

LongText-Bench

Text rendering example: LongText-Bench

LongText-Bench[?] = Long Text-Rendering **Benchmark**

Goal. Render readable multi-line text in images.

Characteristics:

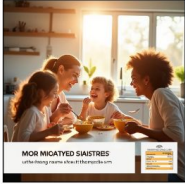
- 160 English + 160 Chinese prompts
- 8 scenarios: signs, labels, printed materials, webpages, slides, posters, captions, dialogues.

OCR match

Evaluation criteria. OCR extraction of rendered text.

Text rendering example: LongText-Bench

Samples.

	Texts				
	"MORNING MOMENTS MATTER", "Our family starts every day with nutritious goodness"	"Home", "Destinations", "Travel Tips", "About Me", "Wander freely, embrace new cultures and create lifelong travel memories"	"软装Tips", "装修这事儿你得知道"	"开业大吉", "花艺坊开业, 欢迎你的到来!", "2022年12月28日 11:00"	
GPT-4o					
BAGEL					
OmniGen2					
X-Omni (ours)					

Common benchmarks

Object alignment

- Verify objects, counts, colors
- Tests compositional image generation
- Fine-grained failure diagnosis

GenEval

Dense prompts

- Follow long detailed prompts
- Track entities, attributes, relations
- Evaluates semantic completeness

DPG-Bench

Text rendering

- Generate readable text in images
- Handles long strings
- Tests OCR-level fidelity

LongText-Bench

Image editing

- Apply user edit instructions
- Preserve source image quality
- Covers real-world edit types

GEdit-Bench

Image editing example: GEdit-Bench

GEdit-Bench [?] = **G**rounded **E**dit **B**enchmark

Goal. Edit an image without destroying the original context.

Characteristics:

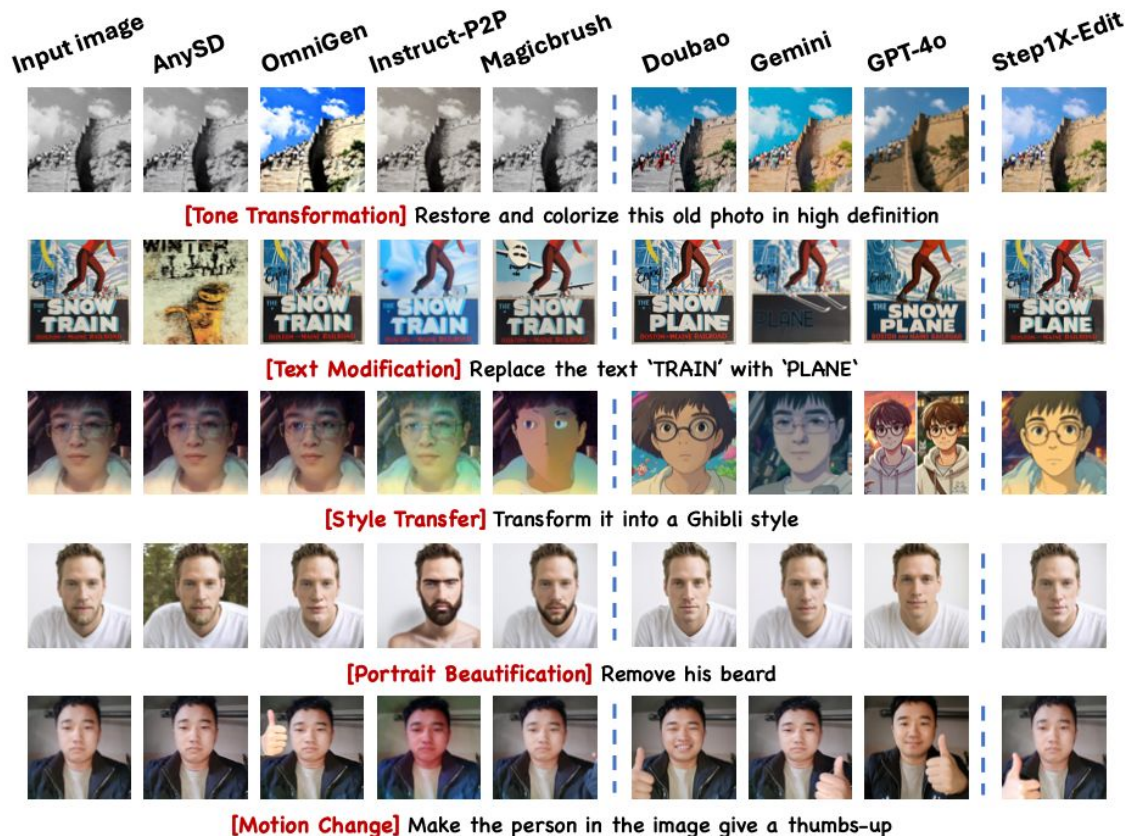
- ~600 editing examples
- 11 categories: background/color/material/motion/style/text change, photoshopping, subject add/remove/replace, tone transfer

Evaluation criteria. VIEScore-style judge.

MLLM-as-a-Judge

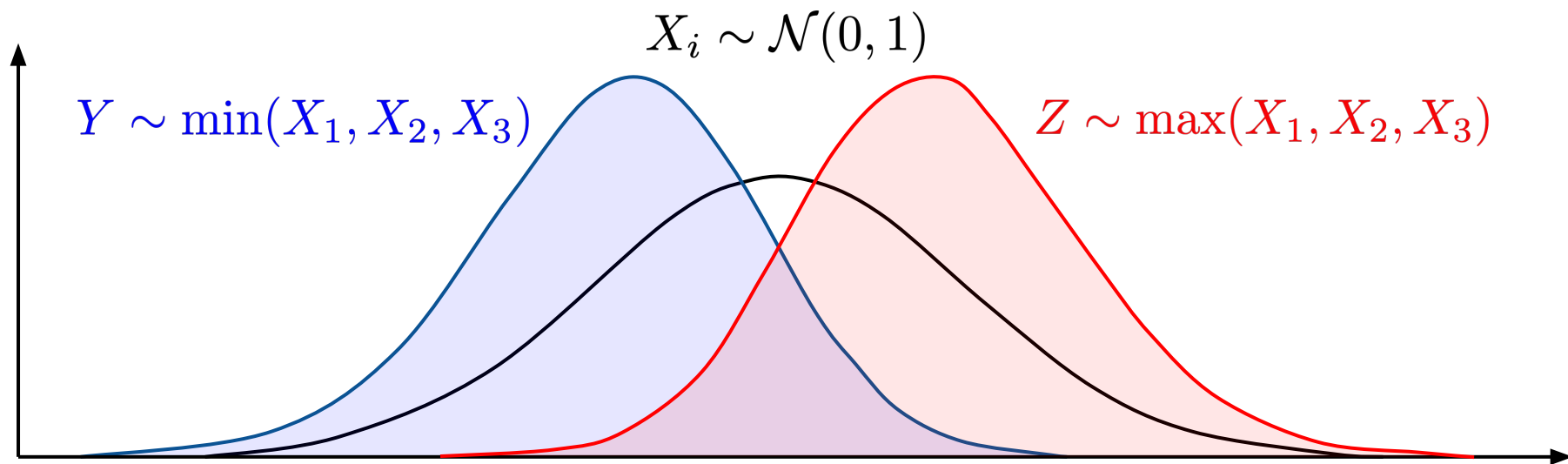
Image editing example: GEdit-Bench

Samples.



Last thoughts

Sample images. "See how well my model performs!"



Beware when used as "proof"!

Thank you for your attention!